

Tally Integration and Business Analytics Platform - SRS v7.3 Complete Edition

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Tally Integration & Business Analytics Platform

Software Requirements Specification — v7.3 Complete Edition

Item	Value
Document	Software Requirements Specification (IEEE 830-style)
Version	7.3 — Complete Edition
Status	Implementation-ready for development, subject to the validation gates in Section 27. Not claimed as production-ready.
Supersedes	All earlier versions (v1-v7.3). This edition is self-contained: no earlier version needs to be read alongside it. Version history is summarized in Appendix A.

Status markers used in this document

Marker	Meaning
LIVE TALLY VERIFICATION REQUIRED	The statement depends on how TallyPrime actually behaves or exports data. It must be confirmed against a real TallyPrime export before it is relied on.
OPEN – BUSINESS DECISION	A stakeholder must decide. A default is given and applies until the decision is made.
IMPLEMENTATION TEST	A value or approach must be confirmed by a prototype or measurement during development. A default is given.

Requirement ID prefixes

Prefix	Area
FR	General functional requirement
AGT / RTE / VER	Tally Sync Agent / command routing / version compatibility
SYNC / VAL	Synchronization / Tally validation gates
DR	Data and record identity
ACC	Accounting and analytics rules
REC	Reconciliation

REC	RECONCILIATION
AGE / FR-AGE / FR-PAY	Aging / payment behaviour
FR-STK	Stock analytics
FR-DD / EXP	Drill-down / exports
RBAC / SEC	Access control / security
LOG	Logging and audit
PERF / NFR / TZ / Q	Non-functional requirements / time zone / quarters
BKP	Backup and recovery
TEST	Testing
AC	Acceptance criterion

1. Introduction

1.1 Purpose

This document specifies a web-based platform that:

1. Extracts accounting, inventory and transaction data from **TallyPrime** into a PostgreSQL database, using custom **TDL (Tally Definition Language)** Collections and Reports invoked over Tally's XML/HTTP interface, through a local **Tally Sync Agent**.
2. Presents a **business analytics dashboard** computed only by direct, deterministic aggregation of the synced data. No AI, ML or LLM component is used in the analytics.
3. Verifies that the local data matches Tally through an independent **reconciliation** process.
4. Optionally offers **anomaly detection**, where anomalies are found by deterministic rules and an LLM (Claude, via an MCP server) may be used only to write a plain-language explanation of evidence the application has already computed.

1.2 Scope

In scope

- TDL-driven, one-way synchronization (Tally → platform) of companies, groups, ledgers, voucher types, stock items, cost centres, opening balances, stock snapshots, and vouchers (header, ledger entries, inventory entries, bill allocations, cost-centre allocations).
- Local Tally Sync Agent (Windows), supporting one or more Agents per company.
- AI-free analytics: sales, purchases, customer and product revenue, expenses, cash flow, balances, receivables/payables aging, customer payment behaviour, stock movement.
- Reconciliation of local data against Tally totals.
- Drill-down from every summary figure to the vouchers behind it; CSV/PDF export.
- Authentication, role-based access control, company (tenant) isolation, audit logging.
- Optional, feature-flagged anomaly detection with optional AI explanation.
- Responsive web interface usable on phone-width screens.

Out of scope

- Writing any data back into Tally (no two-way sync, no data entry, no invoice creation).
- Payroll.
- Forecasting and natural-language querying.
- Replacing Tally as the system of record for statutory compliance.
- GST analytics (default; see OPEN – BUSINESS DECISION in Section 8.15).
- Multi-company consolidated reporting in the user interface (the data model supports multiple companies; the v1 interface works with one company at a time).

1.3 Scope decisions for features found in comparable products

Feature	Decision	Reason
Stock by godown / batch	Out of scope for v1	Adds godown and batch masters and per-location quantities; stock is reported at company level
Order vouchers (Sales/Purchase Order) and inventory-only vouchers (Delivery Note, Receipt Note, Stock Journal)	Not synced in v1	Orders have no accounting effect. Inventory-only vouchers affect stock quantity, so current stock is taken from a Tally-computed snapshot instead (Section 11.3)
Profitability / gross profit	Out of scope for v1	Requires stock valuation (cost of goods sold)
Customer payment behaviour	In scope (Section 10.4)	Reuses bill-allocation data already required for aging
Payment reminders (WhatsApp/SMS/email)	Out of scope	The platform is read-only
Mobile access	In scope as responsive web; native app out of scope	The dashboard must work on phone-width screens
Offline access to synced data	Out of scope	The dashboard requires a connection to the backend

1.4 Definitions

Term	Meaning
TDL	Tally Definition Language, used to define custom Collections and Reports that control exactly what data Tally exports
TDL Collection	A TDL-defined query over Tally objects (Ledger, Voucher, StockItem, etc.)
TDL Report	A TDL definition that formats a Collection's data and returns it as XML over Tally's HTTP interface
Tally XML/HTTP server	Tally's built-in HTTP server (default port 9000) that accepts XML export requests
Tally Sync Agent (Agent)	A small Windows service installed on or near the Tally machine. It talks to Tally locally and to the backend over outbound HTTPS only
Company	One Tally company, identified by its Tally GUID
GUID	Tally's globally unique identifier for an object. The platform's identity key for every synced record
ALTERID	Tally's change counter for an object, used to detect new and modified records
Sync watermark	The highest ALTERID successfully stored for one company and one collection type
Sync lease	A lock that allows only one Agent at a time to sync a given collection for a company
Stale record	An incoming record whose ALTERID is lower than the stored ALTERID for the same GUID
Voucher	A Tally transaction, made of a header, ledger entries, and optionally inventory entries, bill allocations and cost-centre allocations
Ledger entry	One debit or credit line of a voucher, posted to one ledger
Group	A Tally account group. Groups are nested
Primary group	The predefined top-level group a ledger ultimately belongs to, found by walking up the group hierarchy
Base voucher type	The predefined Tally voucher type (Sales, Purchase, Receipt, Payment, Contra, Journal, Credit Note, Debit Note) that a user-created voucher type derives from
Opening balance	A ledger's balance, or a stock item's quantity and value, at the start of a financial year
Stock snapshot	Tally-computed closing quantity of a stock item as of a sync date
Bill allocation	The part of a ledger entry assigned to a bill reference (for bill-wise receivables/payables)
accounting_direction	The normalized debit/credit direction of a ledger entry, independent of Tally's raw XML format
Total Sales Revenue	The authoritative sales figure (Section 8.4)
Customer-attributed Revenue	Sales revenue attributed to a customer when the transaction has exactly one resolvable customer ledger
Unattributed Customer Revenue	Sales revenue with no resolvable customer, or with more than one candidate customer
Product-attributed Revenue	Revenue from inventory line items
Product Attribution Difference	Total Sales Revenue minus Product-attributed Revenue, shown while the two have not been proven to use the same accounting basis
Unattributed / Non-product Sales Revenue	The same difference, shown under this label only after the same-basis proof has passed
Top-N	A ranked subset (default 10) of a list. Never assumed to add up to the total
Financial quarter	A quarter counted from the company's financial-year start date
company_timezone	The IANA time zone of a company, used for all date boundaries
MCP	Model Context Protocol: a standard for exposing application functions as tools to an LLM client
User-defined field (UDF)	A field added to Tally masters or vouchers through a TDL customisation

1.5 References

Reference	Use

Tally Solutions — TDL documentation (Collections, Reports, Forms, Parts, Lines, Fields)	TDL export design
Tally Solutions — XML/HTTP interface documentation	Transport
Model Context Protocol specification and Python SDK	Optional anomaly explanation layer
tally-database-loader (github.com/dhananjay1405/tally-database-loader), MIT licence	Reference implementation: master-data coverage, extraction batch limits, company targeting, known Tally runtime behaviour
Biz Analyst (bizanalyst.in), commercial product	Feature-scope comparison; confirms the desktop-connector architecture

The two reference implementations are guidance only. Nothing learned from them is treated as confirmed Tally behaviour until it passes the validation gate (Section 7).

2. Overall Description

2.1 Product perspective

The platform sits beside Tally and never replaces it. Tally remains the source of truth for the books. The platform keeps a synchronized, read-only copy for analysis, and proves through reconciliation that the copy matches.

The system has two layers:

- **Core layer (always on, no AI):** Agent, synchronization, database, analytics, reconciliation, dashboard. It works fully with no AI service.
- **Optional anomaly layer (off by default):** deterministic anomaly rules, plus an optional Claude explanation through MCP. Turning it off, or losing the AI service, never affects the core layer.

2.2 User classes

User class	Description	Key needs
Owner	Business owner, usually non-technical	Clear dashboard, trustworthy numbers, control of users and connections
Accountant	Manages the books in Tally	Accurate figures, sync status, reconciliation, anomaly review
Admin/Developer	Maintains the integration	Agent management, TDL definitions, logs, configuration

2.3 Operating environment

Component	Environment
TallyPrime	Windows machine, XML/HTTP server enabled, custom TDL files loaded, running in a logged-in user session
Tally Sync Agent	Windows service on the Tally machine or a machine on the same LAN that can reach Tally's port
Backend	Python (FastAPI), containerized (Docker), cloud-hosted
Database	PostgreSQL (production). SQLite is permitted only for isolated local unit tests
Frontend	React single-page application in a modern browser, desktop or phone
Optional	MCP server (Python mcp SDK) and Claude, used only when anomaly detection is enabled

2.4 Design constraints

1. The analytics layer must not call any AI, ML or LLM component. Every figure must be traceable to a deterministic query over synced data.
2. Data extraction uses custom TDL Collections/Reports, not Tally's default reports, so the fields returned are controlled.
3. The Agent initiates every connection. The backend never connects into a customer network.
4. No synced accounting record is ever hard-deleted.
5. No MCP tool may execute arbitrary SQL, write to the database, or write to Tally.
6. Tally provides no push notifications; all synchronization is pull-based.

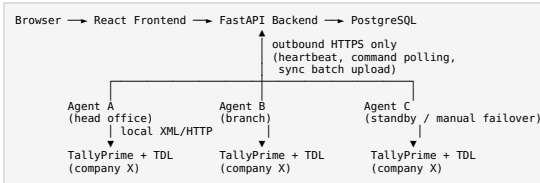
2.5 Assumptions and dependencies

1. The company's chart of accounts is set up in Tally before synchronization.
2. Bill-wise details are maintained in Tally for ledgers where bill-wise aging is expected. Ledgers without them are reported without aging buckets (Section 10.3).
3. Custom TDL files are loaded into Tally on each Agent machine.
4. TallyPrime is kept running in a logged-in Windows session while synchronization is expected.
5. If anomaly explanation is enabled, Claude is reachable through the MCP client.

3. Architecture and Deployment

3.1 Deployment topology

A cloud-hosted backend cannot open a connection to a private address such as 192.168.x.x inside a customer's network. The platform therefore uses a local Agent that makes **outbound HTTPS calls only**. No firewall change, port forwarding or VPN is needed on the customer side.



3.2 Connectivity design rules

Question	Design rule
How does the website reach Tally?	It does not. The Agent reaches the backend over outbound HTTPS
Is a local Agent required?	Yes, for the hosted deployment
Where does the Agent run?	On the Tally machine, or a machine on the same LAN that can reach Tally's XML/HTTP port
How does the Agent talk to Tally?	XML export requests to Tally's HTTP server, using the project's TDL Collections/Reports
How does the Agent talk to the backend securely?	TLS 1.2+ with a per-Agent credential (Section 4.3)
Internet unavailable?	The Agent queues results locally and retries (Section 4.12)
Tally closed?	The Agent reports Tally as unreachable; nothing is lost; sync resumes when Tally is available
Tally XML server disabled?	The Agent reports TALLY_SERVER_DISABLED with an actionable message
How are host and port configured?	Per Agent, by Owner/Admin; default localhost:9000
How is the company selected and identified?	Each Agent is registered to one Tally company, named in every request and verified by GUID on every sync (Sections 4.6, 4.7)
Multiple Tally installations?	One Agent per installation; a company may have several Agents (Section 4.5)

3.3 Direct-connect mode

If the backend runs on infrastructure the business controls on the same network as Tally, the Agent software runs on that backend host and reaches Tally directly. The data model, identity rules and synchronization logic are identical in both modes; only where the Agent runs changes.

4. Tally Sync Agent

4.1 Responsibilities

The Agent (1) registers once with the backend, (2) sends heartbeats and polls for commands, (3) pulls data from Tally using the TDL Collections, (4) uploads data in batches, (5) buffers data locally when the backend is unreachable, and (6) reports Tally and Agent status.

4.2 Registration and onboarding

Step	Action
1	Owner creates a company account in the web application
2	Backend creates a companies row (inactive until an Agent is confirmed)
3	Owner/Admin generates a one-time registration token (valid 24 hours) for that company
4	The customer installs the Agent on the Tally machine and gives it a name (e.g. "Head Office")
5	The customer enters the registration token in the Agent
6	The Agent reads the Tally company's name and GUID from local Tally and sends them with the token
7	Backend validates the token, creates an agent's row bound to the company, stores the Tally GUID and company name, and returns a permanent Agent credential

8	Backend marks the company active (if this is its first Agent) and returns configuration: poll interval, batch size, TDL definitions expected
9	The Agent begins heartbeats
10	Owner/Admin triggers the initial full sync

For a company's first Agent, the Tally GUID it reports becomes the company's `tally_guid`. Every later Agent for the same company must report the same GUID, or registration is rejected with `COMPANY_MISMATCH`.

4.3 Agent authentication

ID	Requirement
SEC-2.0	All Agent-backend traffic uses HTTPS with TLS 1.2 or later
SEC-2.0a	Each Agent holds its own long random credential, sent as a bearer token on every request
SEC-2.0b	The backend stores only a salted hash of each credential and verifies requests by hashing and comparing
SEC-2.4	No HMAC request-signing scheme is used or claimed. The model is bearer credential over TLS only

4.4 Credential rotation and revocation

Step	Rotation
1	Owner/Admin requests rotation for one Agent from the dashboard
2	Backend generates a new credential and invalidates the old hash in the same transaction
3	The new credential is shown once to the requesting user in their authenticated session. It is never pushed to the Agent
4	The user enters the new credential into the Agent's configuration
5	Calls with the old credential are rejected with 401 <code>CREDENTIAL_INVALID</code>

ID	Requirement
SEC-2.1	Rotation invalidates the old credential atomically with issuing the new one. There is no window where both are valid
SEC-2.2	An invalid or revoked credential is rejected with a distinguishable error (<code>CREDENTIAL_INVALID</code> or <code>AGENT_REVOKED</code>)
SEC-2.3	Rotation keeps the same <code>agent_id</code> . Revocation is permanent; the installation must re-register as a new Agent

4.5 Multiple Agents per company

One company can have one or more Agents (branch installations, separate Tally environments, a standby Agent). One Agent serves exactly one Tally company.

Agent state	Meaning
REGISTERING	Token exchanged; awaiting first successful Tally GUID confirmation
ACTIVE	Heartbeating normally
OFFLINE	No heartbeat within the offline threshold (default 5 minutes, <code>IMPLEMENTATION_TEST</code>)
INCOMPATIBLE	Heartbeating, but its version or TDL version is below the backend minimum
REVOKED	Credential revoked; permanently rejected

Standby Agents: there is no automatic failover. If a primary Agent is offline or its command fails, an Owner/Admin sends a new command to the standby Agent. Automatic failover is a possible future feature, not part of this specification.

4.6 Agent ↔ Tally company mapping

ID	Requirement
AGT-3.1	Each Agent records the Tally company GUID it is bound to at registration
AGT-3.2	Before every sync, the Agent confirms that the targeted Tally company's GUID matches its registered GUID
AGT-3.3	On a mismatch the Agent raises <code>COMPANY_MISMATCH</code> , halts, and pulls and uploads nothing

AGT-3.4	A mismatch is shown as a warning against that Agent and requires explicit re-registration. An existing Agent is never silently re-bound to another company
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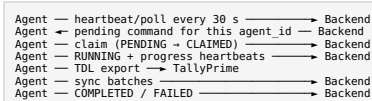
4.7 Explicit company targeting

TallyPrime can have several companies loaded at once. Relying on the active company risks pulling data from the wrong one.

ID	Requirement
AGT-5.1	Every TDL export request names the Agent's registered company. The Agent never relies on Tally's active company
AGT-5.2	The name targets the request; the GUID check (AGT-3.2) confirms the result
AGT-5.3	If the named company is not loaded in Tally, the Agent reports <code>COMPANY_NOT_LOADED</code> and pulls nothing
AGT-5.4	If the company is renamed in Tally, the Agent reports <code>COMPANY_NOT_LOADED</code> ; an Owner/Admin updates the stored name. No re-registration is needed, because the GUID is unchanged
AGT-5.5	The exact request mechanism for naming a company is <code>LIVE TALLY VERIFICATION REQUIRED</code>

4.8 Heartbeat, command polling and recovery

The backend never contacts an Agent. Manual and scheduled syncs reach the Agent as **commands** it collects when it polls.



ID	Requirement
AGT-1.1	The Agent polls at a configurable interval, default 30 seconds. The heartbeat carries Agent version, TDL version, Tally version where available, Tally process uptime and queue status
AGT-1.2	Command states: <code>PENDING</code> → <code>CLAIMED</code> → <code>RUNNING</code> → <code>COMPLETED</code> , or <code>FAILED</code> , <code>EXPIRED</code> , <code>FAILED_AGENT_LOST</code>
AGT-1.3	Claiming is an atomic compare-and-set on the command's status, so a command can be claimed only once
AGT-1.4	A failed command is not retried automatically; a new command must be issued
AGT-1.5	"Sync Now" creates a <code>PENDING</code> command; it never attempts to connect to the Agent
AGT-1.6	Command status is visible on the dashboard as it changes
AGT-1.7	A <code>RUNNING</code> command has a lease deadline (<code>lease_expires_at</code>) that the Agent extends with progress heartbeats
AGT-1.8	If the lease deadline passes, the command becomes <code>FAILED_AGENT_LOST</code> and the reason is recorded
AGT-1.9	A <code>FAILED_AGENT_LOST</code> or <code>EXPIRED</code> command is never reassigned to another Agent
AGT-1.10	A <code>PENDING</code> command not claimed within a configurable window (default 10 minutes) becomes <code>EXPIRED</code>

4.9 Command routing

ID	Requirement
RTE-1.1	If a company has exactly one <code>ACTIVE</code> Agent, "Sync Now" targets it automatically
RTE-1.2	If it has more than one, the user must choose an Agent; the backend rejects a request without <code>agent_id</code>
RTE-1.3	Every scheduled sync is bound to a specific Agent when the schedule is created
RTE-1.4	A command's <code>agent_id</code> is mandatory and cannot change after creation
RTE-1.5	The backend verifies that the Agent belongs to the requesting user's authorized company. Commands never cross company boundaries
RTE-1.6	A command for an <code>OFFLINE</code> Agent stays <code>PENDING</code> , labelled "Waiting — Agent offline since ", and is claimed when the Agent returns, unless it expires first

4.10 Tally-side extraction batch size

ID	Requirement
AGT-4.1	The Agent requests vouchers from Tally in batches; default 5,000 vouchers per request, configurable per Agent
AGT-4.2	Batch sizes above 10,000 are rejected
AGT-4.3	A Tally request with no response within a timeout (default 10 minutes) is abandoned, logged as TALLY_EXPORT_TIMEOUT, and retried once at half the batch size; if it fails again the sync segment fails
AGT-4.4	These defaults come from the reference implementation and are confirmed on the project benchmark machine (IMPLEMENTATION TEST)

4.11 Tally runtime operating constraints

ID	Requirement
AGT-6.1	TallyPrime must be running in a logged-in Windows session. The Agent cannot start or keep Tally running
AGT-6.2	Installation guidance tells Remote Desktop users to disconnect, not log off, because logging off closes Tally
AGT-6.3	The Agent reports Tally's process uptime, read from the Windows process table
AGT-6.4	When uptime exceeds a threshold (default 7 days), the dashboard advises restarting Tally, more prominently if the latest reconciliation found a discrepancy

4.12 Local queue

ID	Requirement
AGT-2.1	Each Agent keeps a local queue (a local SQLite file) of batches waiting to upload
AGT-2.2	Queue limit: 50,000 records or 7 days of data, whichever comes first (configurable)
AGT-2.3	Upload retries use exponential backoff from 30 seconds up to 15 minutes
AGT-2.4	When the queue is full, the Agent stops pulling new data, keeps retrying uploads, and reports QUEUE_FULL
AGT-2.5	A batch that fails upload 20 times (configurable) moves to a local dead-letter file and is reported; it is never silently dropped
AGT-2.6	The queue file is restricted by operating-system permissions to the Agent's service account. Disk encryption is the customer's responsibility

4.13 Version compatibility

ID	Requirement
VER-1.1	The Agent reports its version, TDL package version and, where obtainable, TallyPrime version on every heartbeat
VER-1.2	The backend holds a minimum compatible Agent version and TDL version. An Agent below either is marked INCOMPATIBLE and receives no commands
VER-1.3	Certified TallyPrime releases are recorded in a compatibility table as they are tested (LIVE TALLY VERIFICATION REQUIRED); no version numbers are assumed in advance

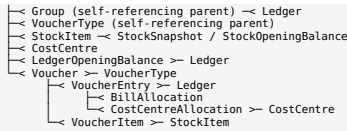
5. Data Model

5.1 Principles

1. Every company-scoped table carries company_id, even though the v1 interface works with one company at a time. Adding a company is a new row, not a schema change.
2. Every synced master and voucher is identified by (company_id, tally_guid). Voucher numbers are for display only.
3. Every synced master and voucher stores its latest alter_id and a lifecycle status.
4. Accounting amounts are stored in a normalized form that does not depend on Tally's raw XML sign format.
5. No synced accounting record is hard-deleted.

5.2 Entity relationships

Company  Agent  AgentCommand



5.3 Company, users and access

Table	Key fields	Constraints / indexes
companies	company_id (UUID PK), tally_guid, name, financial_year_start, company_timezone (IANA), is_active, created_at	UNIQUE(tally_guid)
users	user_id PK, email, password_hash (bcrypt), name, is_active, created_at	UNIQUE(email)
roles	role_id PK, role_name (OWNER, ACCOUNTANT, ADMIN)	UNIQUE(role_name)
user_roles	user_id, company_id, role_id	PK(user_id,company_id,role_id); index company_id

5.4 Agents, commands and schedules

Table	Key fields	Constraints / indexes
agents	agent_id PK, company_id, agent_name, credential_hash, status, tally_guid, tally_company_name, tally_host, tally_port, extraction_batch_size, agent_version, tdl_version, tally_version, tally_uptime_seconds, queue_status, last_heartbeat_at, registered_at, revoked_at	UNIQUE(company_id,agent_name); index company_id
agent_registration_tokens	token_id PK, company_id, token_hash, expires_at, used_at, created_by	index company_id
agent_commands	command_id PK, company_id, agent_id (mandatory, immutable), command_type (RUN_SYNC), sync_mode (FULL/INCREMENTAL/DATE_RANGE/RECONCILIATION), date_from, date_to, status (PENDING/CLAIMED/RUNNING/COMPLETED/FAILED/EXPIRED/FAILED_AGENT_LOST), lease_expires_at, created_by, created_at, claimed_at, completed_at, error_message	index (agent_id,status)
sync_schedules	schedule_id PK, company_id, agent_id, cron_expression, sync_mode, is_active, created_by	index company_id

5.5 Masters

Table	Key fields	Constraints / indexes
groups	group_id PK, company_id, tally_guid, name, parent_group_id (null for a primary group), primary_group_id, nature (ASSET/LIABILITY/INCOME/EXPENSE), alter_id, status	UNIQUE(company_id,tally_guid); index primary_group_id
ledgers	ledger_id PK, company_id, tally_guid, name, group_id, primary_group_id (cached), alter_id, status, custom_fields (JSONB), last_synced_at	UNIQUE(company_id,tally_guid); index (company_id,primary_group_id)
voucher_types	voucher_type_id PK, company_id, tally_guid, name, parent_voucher_type_id, base_voucher_type (SALES/PURCHASE/RECEIPT/PAYMENT/CONTRA/JOURNAL/CREDIT_NOTE/DEBIT_NOTE/OTHER), alter_id, status	UNIQUE(company_id,tally_guid)
stock_items	stock_item_id PK, company_id, tally_guid, name, base_unit, gst_rate (descriptive only), alter_id, status, custom_fields (JSONB)	UNIQUE(company_id,tally_guid)
cost_centres	cost_centre_id PK, company_id, tally_guid, name, alter_id, status	UNIQUE(company_id,tally_guid)

ledgers has no stored closing balance. Tally has no fixed closing-balance field, because a balance depends on the date; balances are computed (Section 8.10).

5.6 Opening balances and stock snapshots

Table	Key fields	Primary key
ledger_opening_balances	company_id, ledger_id, financial_year_start, amount_absolute, accounting_direction	(ledger_id,financial_year_start)
stock_opening_balances	company_id, stock_item_id, financial_year_start, quantity, unit, value	(stock_item_id,financial_year_start)
stock_snapshots	company_id, stock_item_id, as_of_date, closing_quantity, unit, sync_run_id	(stock_item_id,as_of_date)

5.7 Vouchers

Table	Key fields	Constraints / indexes
vouchers	voucher_id PK, company_id, tally_guid, voucher_number (display only), voucher_type_id, voucher_date, narration, status (ACTIVE/CANCELLED/MISSING_IN_TALLY), alter_id, custom_fields (JSONB), last_synced_at	UNIQUE(company_id,tally_guid); index (company_id,voucher_date); index (company_id,alter_id)
voucher_entries	voucher_entry_id PK, voucher_id, ledger_id, line_sequence, stable_line_id (nullable, Section 6.9), amount_raw, is_debit, amount_absolute, amount_signed, accounting_direction (DEBIT/CREDIT)	index voucher_id; index ledger_id
bill_allocations	bill_allocation_id PK, voucher_entry_id, allocation_type, reference_name, due_date (nullable), amount_absolute	index voucher_entry_id; index (reference_name); index due_date
cost_centre_allocations	id PK, voucher_entry_id, cost_centre_id, amount_absolute	index voucher_entry_id
voucher_items	voucher_item_id PK, voucher_id, stock_item_id, quantity, unit, rate, amount, custom_fields (JSONB)	index voucher_id; index stock_item_id

5.8 Normalized accounting fields

Field	Meaning
amount_raw	Value exactly as received from Tally, kept for audit
is_debit	Set by the parser from Tally's debit/credit indicator
amount_absolute	Always non-negative
amount_signed	Debit positive, credit negative, applied uniformly by the parser
accounting_direction	DEBIT or CREDIT; the field all analytics read

ID	Requirement
ACC-DATA-1	All analytics use accounting_direction and amount_absolute (or amount_signed), never amount_raw
ACC-DATA-2	Conversion from Tally's raw format to the normalized fields lives in one parser module with its own unit tests
ACC-DATA-3	The exact Tally XML indicator of debit/credit is LIVE TALLY VERIFICATION REQUIRED

5.9 Synchronization control

Table	Key fields	Notes
sync_watermarks	PK(company_id,collection_type); last_alter_id, last_successful_sync_at, status, locked_by_agent_id, lock_acquired_at, lock_expires_at	collection_type: COMPANY, GROUP, LEDGER, VOUCHER_TYPE, STOCK_ITEM, COST_CENTRE, VOUCHER. The lock fields implement the sync lease (Section 6.3)
sync_runs	sync_run_id PK, company_id, agent_id, command_id, sync_mode, started_at, ended_at, status (IN_PROGRESS/COMPLETED/FAILED/PARTIAL), records_fetched, records_failed	index (company_id,started_at)
sync_errors	sync_error_id PK, sync_run_id, entity_type, tally_guid, error_code, message, created_at	index sync_run_id
reconciliation_results	result_id PK, company_id, run_at, metric, entity_id (nullable, e.g. ledger), period_start, period_end, tally_value, local_value, absolute_difference, percentage_difference, result (PASS/FAIL)	index (company_id,run_at)

5.10 Configuration, audit and optional anomaly tables

Table	Key fields
feature_config	PK(company_id,feature_name); enabled (boolean only), updated_by, updated_at
company_settings	PK(company_id,setting_key); setting_value, data_type (INTEGER/DECIMAL/BOOLEAN/JSON), updated_by, updated_at

custom_field_mappings	company_id, collection_type, tally_field, field_key, data_type, is_active, updated_by, updated_at
audit_logs	audit_log_id, company_id, user_id (null for system), action, entity_type, entity_id, before_value, after_value, data_range, result, created_at — append-only
anomaly_flags	anomaly_id, company_id, voucher_id, rule_triggered, transaction_amount, historical_average, historical_max, deviation_percent, duplicate_of_voucher_id, flagged_at, explanation_text, explanation_status (AVAILABLE/UNAVAILABLE/PENDING), reviewed, not_an_issue, reviewed_by, reviewed_at
ai_tool_log	id, company_id, tool_name, parameters, result_summary, status, created_at

5.11 User-defined fields

ID	Requirement
DR-UDF-1	No user-defined fields are extracted by default; Owners/Admins add mappings
DR-UDF-2	Mapped values are stored in custom_fields, shown in drill-down and included in exports, and never used in analytics calculations
DR-UDF-3	A mapped field missing from an export is stored as null and logged once per run as UDF_NOT_FOUND; the sync continues
DR-UDF-4	The TDL export field list is generated from custom_field_mappings, not edited by hand

6. Synchronization

6.1 Sync modes

Mode	Behaviour
Initial full sync	Pulls every record of every collection with no ALTERID filter and sets the watermarks
Incremental sync	Pulls records with ALTERID greater than the collection's watermark. Default schedule hourly
Manual sync	"Sync Now" creates a command (Section 4.8); runs incremental by default
Date-range sync	Manual, for re-verifying a period. Does not move a watermark unless it processes records beyond it
Reconciliation sync	Daily by default: pulls totals and key lists and compares them with local data (Section 9)

6.2 Watermarks

ID	Requirement
SYNC-1.1	Each (company_id, collection_type) has its own watermark. Collections are synced as independent streams
SYNC-1.2	A watermark advances only after the batch containing those records has been committed
SYNC-1.3	The shared per-company watermark is valid only while all of a company's Agents connect to the same Tally company GUID. Consolidating separate Tally companies under one business account is outside this specification

6.3 Sync lease

ID	Requirement
SYNC-4.1	Before syncing a collection, an Agent acquires the lease on its sync_watermarks row with an atomic compare-and-set
SYNC-4.2	If another Agent holds an unexpired lease, the request is rejected with SYNC_LOCKED and nothing is pulled for that collection
SYNC-4.3	The lease is released on completion, or reclaimed automatically after lock_expires_at
SYNC-4.4	The lease prevents concurrent writers; stale-record protection (Section 6.4) prevents out-of-order data. Both are required

6.4 Stale-record protection

ID	Requirement
SYNC-3.1	Every upsert compares the incoming ALTERID with the stored ALTERID for that (company_id, GUID)
SYNC-3.2	Incoming lower than stored: the record is rejected, the local record is unchanged, and STALE_ALTERID is logged
SYNC-3.3	Equal: no change, not an error
SYNC-3.4	Higher: the update is applied (Section 6.9 for vouchers)
SYNC-3.5	This rule applies to groups, ledgers, voucher types, stock items, cost centres and vouchers

6.5 Batching, idempotency and partial failure

ID	Requirement
SYNC-6.1	The backend commits uploads in batches (default 500 records), each in its own transaction
SYNC-6.2	Each voucher's full write (header, entries, bill allocations, cost-centre allocations, items) is one transaction; a voucher is never left partly written
SYNC-6.3	Every write is an upsert keyed on (company_id, tally_guid), so replaying a batch is always safe
SYNC-6.4	If a run fails part-way (for example after 50,000 of 100,000 records), committed batches remain, the watermark reflects the last committed batch, the run is marked PARTIAL, and the next run resumes from the watermark with no duplicates
SYNC-6.5	A record that cannot be parsed is logged to sync_errors; the rest of the batch continues
SYNC-6.6	If Tally is unreachable, no local data is changed

6.6 Record identity

ID	Requirement
DR-4.1	A voucher is identified by (company_id, tally_guid). voucher_number is display-only, because Tally voucher numbers can repeat across financial years and be edited
DR-4.2	Groups, ledgers, voucher types, stock items and cost centres are also identified by (company_id, tally_guid)
DR-4.4	Two companies using the same voucher number, or a reused voucher number, cannot cause a collision
DR-4.6	UNIQUE(company_id,tally_guid) is enforced by the database on every synced master and voucher table

6.7 Voucher lifecycle

State change	Detection	Local effect
New	GUID not stored	Insert; status = ACTIVE
Modified	ALTERID higher than stored	Header updated, children replaced (Section 6.9), before/after of header values written to audit_logs
Cancelled	ALTERID higher, cancellation indicator set	status = CANCELLED; row kept; excluded from standard analytics
Deleted	GUID missing from a key list or full reconciliation (Section 6.8)	status = MISSING_IN_TALLY; row kept; excluded from standard analytics; listed for review
Reappeared	A MISSING_IN_TALLY GUID is present again	status = ACTIVE; logged to audit_logs

Worked example: voucher 100 is ₹10,000 on day 1 and edited to ₹15,000 on day 2. Editing raises its ALTERID, so the next incremental sync returns it; the stored ALTERID is lower, so the voucher is updated and the change from ₹10,000 to ₹15,000 is recorded in audit_logs.

6.8 Deletion detection

ID	Requirement
SYNC-5.1	After each incremental sync of a collection, the Agent requests a key-only list (GUID, ALTERID) of that collection for the synced period
SYNC-5.2	Any ACTIVE local record missing from the list becomes

	MISSING_IN_TALLY
SYNC-5.3	The daily full reconciliation remains required; it checks totals and covers periods outside the incremental window
SYNC-5.4	If the key list proves too costly on large companies, it runs on an interval (default hourly) instead (IMPLEMENTATION TEST)
SYNC-5.5	The reference implementation's deletion handling is reviewed before this mechanism is finalized

6.9 Voucher child-row updates

Whether Tally exposes a stable identifier for each ledger-entry and inventory line is LIVE TALLY VERIFICATION REQUIRED.

ID	Requirement
DR-VE-1	A live export is inspected during implementation to determine whether a stable per-line identifier exists
DR-VE-2	If it exists, lines are updated individually using it
DR-VE-3	If not, a modified voucher's children are replaced in full; fragile composite keys (ledger + amount + position) are never used for matching
DR-VE-4	The replacement is one transaction; any failure rolls back the whole voucher

Replacement order, inside one transaction:

1. Update the voucher header.
2. Delete bill allocations and cost-centre allocations belonging to the voucher's ledger entries.
3. Delete the voucher's ledger entries.
4. Delete the voucher's inventory items.
5. Insert the current ledger entries, bill allocations, cost-centre allocations and inventory items.
6. Commit.

6.10 Master-data lifecycle

ID	Requirement
DR-ML-1	Groups, ledgers, voucher types, stock items and cost centres are never hard-deleted by sync
DR-ML-2	Historical vouchers keep their links to a master that becomes MISSING_IN_TALLY; past-period analytics are unaffected
DR-ML-3	MISSING_IN_TALLY masters appear in a "missing masters" review list
DR-ML-4	A master that reappears returns to ACTIVE, and this is logged
DR-ML-5	An INACTIVE state is recorded if Tally exposes one (LIVE TALLY VERIFICATION REQUIRED)

6.11 TDL export definitions

ID	Requirement
FR-1.1	Custom TDL Collections exist for company, groups, ledgers (with opening balances), voucher types, stock items (with opening balances), cost centres, vouchers (header, ledger entries, inventory entries, bill allocations, cost-centre allocations), stock closing quantities as of a date, ledger closing balances as of a date, and reconciliation totals
FR-1.2	Every Collection returns GUID and ALTERID and supports an ALTERID > filter and a key-only form
FR-1.3	A custom TDL Report per Collection returns the data as XML over Tally's HTTP interface
FR-1.4	If a required TDL definition is not loaded in Tally, the Agent reports TDL_NOT_LOADED and never falls back to default reports

7. Tally Validation Gate

The design depends on specific Tally behaviour. Each item below must reach PASSED, with evidence, on a live TallyPrime installation before the dependent feature is declared complete.

ID	Requirement
VAL-1.1	No release claims incremental-sync compatibility for a collection until all of that collection's ALTERID rows are PASSED
VAL-1.2	If an ALTERID row is FAILED, the affected collection falls back to full-pull-only sync

VAL-1.3	This table is a maintained project artifact, updated with evidence during implementation		
#	Test	Expected result	Status
G1	ALTERID on Ledger export	Present, integer	NOT TESTED
G2	ALTERID on Stock Item export	Present, integer	NOT TESTED
G3	ALTERID on Voucher export	Present, integer	NOT TESTED
G4	ALTERID on Cost Centre export	Present, integer	NOT TESTED
G5	ALTERID type and range	Integer, increasing per company	NOT TESTED
G6	ALTERID ordering	Increases on every change to an object	NOT TESTED
G7	ALTERID > filter in TDL	Collection accepts a numeric filter	NOT TESTED
G8	Modified voucher	Editing raises its ALTERID	NOT TESTED
G9	Cancelled voucher	Cancelling raises ALTERID and sets a detectable indicator	NOT TESTED
G10	Consistency across collections	Same ALTERID behaviour for all collection types	NOT TESTED
G11	Deletion	Deleting a record produces no ALTERID signal on other records	NOT TESTED
G12	Group parent reference	Each group's parent is identifiable	NOT TESTED
G13	Primary group resolution	Every group's chain reaches a predefined primary group	NOT TESTED
G14	Group nature	Income/expense/asset/liability nature identifiable from export or fixed mapping	NOT TESTED
G15	Voucher type parent	Each voucher type's parent is identifiable	NOT TESTED
G16	Ledger opening balance	Exportable with debit/credit direction	NOT TESTED
G17	Stock opening balance	Opening quantity and value exportable	NOT TESTED
G18	Stock closing quantity as of a date	Returned for a given date	NOT TESTED
G19	Ledger closing balance as of a date	Returned for a given date	NOT TESTED
G20	Named-company targeting	A request can target a loaded, non-active company	NOT TESTED
G21	Key-only export	Collection exportable with only GUID and ALTERID	NOT TESTED
G22	User-defined field extraction	A TDL-customised field can be requested	NOT TESTED
G23	Debit/credit indicator	The raw indicator is identified and the parser maps it correctly (ACC-DATA-3)	NOT TESTED
G24	Stable line identifier	Whether ledger-entry/inventory lines have a stable identifier (DR-VE-1)	NOT TESTED
G25	Bill allocation types	Exact exported values for New Reference, Against Reference, Advance, On Account	NOT TESTED
G26	Return linkage	How Credit/Debit Notes reference original bills in the export	NOT TESTED
G27	Unit data	Whether base units and conversion factors are exported	NOT TESTED
G28	Product revenue basis	Whether product-line revenue uses the same basis as Total Sales (ACC-VAL-1)	NOT TESTED
G29	Master inactive state	Whether Tally exposes an inactive state for masters	NOT TESTED

8. Accounting and Analytics Rules

8.1 Core principles

1. Every metric is computed by direct SQL/pandas aggregation over synced data. No AI, ML or LLM component is used.
2. Every metric filters ledger entries by **both** the ledger's primary group and the entry's accounting direction. A voucher's entries are never summed as a whole, because a balanced voucher's debits and credits cancel out.
3. Standard analytics include only ACTIVE vouchers. CANCELLED and MISSING_IN_TALLY vouchers are excluded unless the user explicitly asks to see them.
4. All date grouping uses company_timezone (Section 17.5).

8.2 Ledger classification by primary group

ID	Requirement
ACC-7.1	Every rule that classifies a ledger (sales, purchase, expense, cash, bank, duties and taxes, customer, supplier) uses the ledger's primary group, never its immediate parent
ACC-7.2	The primary group is resolved at sync time by following parent groups to a predefined primary group, and cached
ACC-7.3	Classification allow-lists in company_settings contain primary groups only
ACC-7.4	A group whose chain does not reach a primary group (missing parent or a loop) is marked UNRESOLVED_GROUP; its ledgers are excluded from classified metrics and listed for data-quality review
ACC-7.5	When a group's parent changes, primary groups are recalculated for it and everything beneath it on the next sync, and the change is logged
ACC-7.6	Customers are ledgers whose primary group is Sundry Debtors; suppliers are ledgers whose primary group is Sundry Creditors

Default allow-lists: Sales = Sales Accounts; Purchase = Purchase Accounts; Expense = Direct Expenses, Indirect Expenses; Cash/Bank = Cash-in-Hand, Bank Accounts; Tax = Duties & Taxes.

8.3 Voucher classification by base voucher type

ID	Requirement
ACC-8.1	Every rule that filters by voucher type uses base_voucher_type, never the voucher-type name
ACC-8.2	A user-created type such as "POS Invoice" derived from Sales is included wherever Sales is
ACC-8.3	A voucher whose type cannot be resolved is classified OTHER, excluded from type-filtered metrics, and listed for data-quality review

8.4 Sales revenue

ID	Metric	Rule
ACC-1.1	Total Sales Revenue (authoritative)	Sum of CREDIT entries on ledgers whose primary group is in the Sales allow-list, on ACTIVE vouchers whose base type is SALES, minus linked Sales Returns (Section 8.7). In taxable-value mode (default), entries on Duties & Taxes ledgers are excluded
ACC-2.1	No double counting	No metric adds both the debit and credit sides of the same voucher
ACC-2.2	Tax exclusion	Tax-classified entries are excluded from all revenue and purchase metrics in taxable-value mode

8.5 Customer attribution

ID	Requirement
ACC-6.1	A qualifying Sales voucher with exactly one customer ledger among its entries: its revenue is Customer-attributed Revenue for that customer
ACC-6.2	No customer ledger (for example a cash sale): Unattributed Customer Revenue
ACC-6.3	More than one customer ledger: Unattributed Customer Revenue. Revenue is never split between customers by guesswork
ACC-6.4	All customer-attributed revenue plus Unattributed Customer Revenue equals Total Sales Revenue for the same period and filters
ACC-1.4	Supplier amounts follow the same pattern using Purchase Value and Sundry Creditors

8.6 Product attribution

ID	Requirement
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ACC-1.8	Product-attributed Revenue = sum of voucher_items.amount on vouchers that qualify under ACC-1.1
ACC-1.10	By default, Total Sales Revenue minus Product-attributed Revenue is shown as Product Attribution Difference , a data-quality figure, not an accounting claim
ACC-1.9	Only after validation gate G28 passes (Product-attributed Revenue proven to use the same basis as Total Sales: same vouchers, tax treatment, return treatment, discounts and period) is the difference shown as Unattributed / Non-product Sales Revenue
ACC-VAL-1	The dashboard shows exactly one of the two labels above, never both

8.7 Sales and purchase returns

ID	Requirement
ACC-5.1	A Credit Note is a Sales Return only when its bill allocation links it to an original Sales bill reference
ACC-5.2	A Debit Note is a Purchase Return only under the equivalent link to an original Purchase bill
ACC-5.3	Notes without a resolvable link are classified UNLINKED_CREDIT_NOTE / UNLINKED_DEBIT_NOTE
ACC-5.4	Unlinked notes are not subtracted from sales or purchases; they appear in an “Unclassified Adjustments” view
ACC-5.5	How return links appear in the export is gate G26 (LIVE TALLY VERIFICATION REQUIRED); until it passes, all notes are treated as unlinked

8.8 Purchases and expenses

ID	Metric	Rule
ACC-1.2	Purchase Value	Sum of DEBIT entries on ledgers in the Purchase allow-list, on ACTIVE vouchers of base type PURCHASE, minus linked Purchase Returns, tax entries excluded in taxable-value mode
ACC-1.5	Expenses	Sum of DEBIT entries on ledgers in the Expense allow-list, grouped by ledger, primary group or cost centre

8.9 Cash flow

ID	Requirement
ACC-1.6	Cash inflow = DEBIT entries on Cash/Bank ledgers in vouchers of base type RECEIPT
ACC-1.7	Cash outflow = CREDIT entries on Cash/Bank ledgers in vouchers of base type PAYMENT
ACC-3.1	Inflow is positive and outflow negative in the net cash-flow series
ACC-3.3	CONTRA vouchers (transfers between the company’s own cash and bank ledgers) are excluded
ACC-3.4	JOURNAL vouchers touching cash/bank ledgers are excluded by default and included only if enabled per company (OPEN – BUSINESS DECISION)

8.10 Balances

ID	Requirement
ACC-9.1	A balance-sheet ledger’s balance on date D = its opening balance for the financial year containing D, plus the net of its ACTIVE entries from the start of that year to D
ACC-9.2	Income and expense ledgers report period movements only; their balances do not carry into the next financial year
ACC-9.3	Cash and bank position on date D = sum of balances of ledgers in the Cash/Bank allow-list. Bank overdraft ledgers are included only if added to the list
ACC-9.4	Total receivables and payables on date D use the same rule for Sundry Debtors and Sundry Creditors
ACC-9.5	Computed ledger balances are checked against Tally’s own closing balances on the sync date (Section 9)

ACC-9.6	A ledger with no opening balance for the relevant year shows “opening balance unavailable”; zero is never assumed
---------	---

8.11 Top-N rankings

ID	Requirement
TOPN-1.1	Default N is 10, configurable per company
TOPN-1.2	Top-N lists use exactly the same query as the full list; only the cut-off differs
TOPN-1.3	Each Top-N view is labelled “Top N” and offers “View All”
TOPN-1.4	Top-N totals are never presented as adding up to the overall total

8.12 Cross-module consistency

Metric	Source	Reconciles to
Total Sales Revenue	ACC-1.1	Reference figure
Customer-attributed + Unattributed Customer Revenue	ACC-6.x	Total Sales Revenue
Product-attributed Revenue + Product Attribution Difference (or Unattributed/Non-product Revenue)	ACC-1.8–1.10	Total Sales Revenue
Top-N lists	Same queries, limited	Not required to equal any total

ID	Requirement
ACC-4.4	Dashboard views, drill-downs and exports of a metric all call the same shared query function
ACC-4.5	Cancelled and MISSING_IN_TALLY records are excluded from every standard metric consistently
ACC-4.6	Anomaly evidence is never used as a source for any accounting figure

8.13 Customer-level and product-level views

ID	Requirement
FR-2.4	Top customers are ranked by Customer-attributed Revenue; top products by Product-attributed Revenue and by quantity sold
FR-2.1	Sales, purchases and cash flow can be viewed daily, monthly and by financial quarter

8.14 Configurable accounting settings

The Sales, Purchase, Expense, Cash/Bank and Tax allow-lists, taxable-value mode and journal inclusion are stored in `company_settings` (Section 18).

8.15 GST and tax scope

OPEN – BUSINESS DECISION. **Default: out of scope for v1.** Revenue and purchases are reported at taxable value, excluding tax ledgers. If GST analytics are brought into scope, a `tax_details` table (CGST/SGST/IGST/cess, rate, taxable value, tax amount, HSN/SAC), matching TDL extensions and new requirements must be added. This must be decided before the voucher TDL Collection is finalized.

9. Reconciliation

9.1 Purpose

Reconciliation answers “does the local database match Tally?” using totals from a separate TDL Totals Collection, not by re-adding the synced detail rows. This way a bug in the sync or parser cannot hide itself.

ID	Requirement
REC-1.1	For each company and period, compare Tally and local values for: Sales, Purchases, Receipts, Payments, closing stock quantity per item, and ledger closing balances
REC-1.2	Show the Tally value, the local value, the absolute difference and the percentage difference
REC-1.3	Record every comparison in <code>reconciliation_results</code>
REC-1.4	Run automatically after every full sync and daily reconciliation sync, and on demand
REC-1.5	A voucher or master found only locally during reconciliation is marked

9.2 Tolerance rule

```
absolute_difference = |tally_value - local_value|
percentage_difference = absolute_difference / |tally_value| × 100    (tally_value ≠ 0)

If tally_value = 0: PASS when local_value = 0, otherwise FAIL.
Otherwise:          PASS when absolute_difference ≤ absolute_tolerance
                   OR percentage_difference ≤ percentage_tolerance.
                   FAIL only when both tolerances are exceeded.
```

Metric	Tolerances used	Defaults
Sales, Purchases, Receipts, Payments, ledger balances	Money	₹1.00 absolute, 0.01%
Stock quantity	Quantity	0 units absolute, 0.5%

9.3 Worked examples (₹1 / 0.01%)

Tally	Local	Abs. diff	% diff	Result	Reason
₹100,000	₹100,000.50	₹0.50	0.0005%	PASS	Within absolute tolerance
₹100,000	₹100,010	₹10	0.01%	PASS	Equal to percentage tolerance
₹100,000	₹100,012	₹12	0.012%	FAIL	Both tolerances exceeded
₹100,000	₹100,500	₹500	0.5%	FAIL	Both tolerances exceeded
₹0	₹0	₹0	—	PASS	Zero matches zero
₹0	₹50	₹50	—	FAIL	Non-zero against zero

10. Aging and Payment Behaviour

10.1 Aging categories

Category	Rule
Not yet due	due_date after today
Overdue 0-30 / 31-60 / 61-90 / 90+ days	today - due_date, for due_date on or before today; boundaries configurable, default [30, 60, 90]
Due date unavailable	No due date on the bill

“Today” is the current date in company_timezone. A bill not yet due is never shown with negative overdue days.

10.2 Bill allocation rules

Allocation type	Creates outstanding	Settles outstanding	In aging buckets	Effect
New Reference	Yes	No	Yes	Raises a bill
Against Reference	No	Yes	Reduces the referenced bill	Reduces that bill's outstanding
Advance	No	No	No — shown as “Unadjusted Advances”	Reduces the party's net exposure until allocated to a bill
On Account	No	No	No — shown as “On-Account / Unallocated”	Reduces the party's net exposure until allocated to a bill

ID	Requirement
FR-AGE-1	Aging is computed per bill reference, not per voucher
FR-AGE-2	A bill's outstanding = its New Reference amount minus all Against Reference amounts for the same reference, including those from receipts, payments and linked credit/debit notes
AGE-BILL-1	An allocation type that cannot be mapped to the table above is UNSUPPORTED_ALLOCATION_TYPE
AGE-BILL-2	Unsupported allocations are never included in aging; they are listed for review
AGE-BILL-3	Advance and On Account totals are shown as separate summary lines
AGE-BILL-4	Customer (receivable) and supplier (payable) advances and on-account amounts are shown separately and never netted together

AGE-BILL-5	The exact exported allocation-type values are gate G25 (LIVE TALLY VERIFICATION REQUIRED)
------------	---

10.3 Ledgers without bill-wise details

If bill-wise details are not maintained for a ledger in Tally, its outstanding is shown as one total with the note “bill details not available”. It is never placed into guessed buckets.

10.4 Customer payment behaviour

ID	Requirement
FR-PAY-1	Average days to pay = $\Sigma(\text{settled amount} \times \text{days from bill date to settlement date}) \div \Sigma(\text{settled amount})$, over settlements in a trailing window (default 365 days)
FR-PAY-2	Bill date is the date of the New Reference voucher; settlement date is the date of each Against Reference voucher. Part settlements count separately
FR-PAY-3	Average days past due uses due date instead of bill date; early payments count as 0
FR-PAY-4	Advance and On Account amounts are excluded
FR-PAY-5	Customers with fewer than 3 settlements in the window (configurable) show “insufficient history”
FR-PAY-6	The metric is hidden until gate G25 passes

11. Stock Analytics

11.1 Movement classification

All thresholds are configurable per company. The values shown are defaults.

ID	Class	Rule
FR-STK-1	Measurement period	90 days (30/60/90/180 selectable)
FR-STK-2	Fast-moving	Sales quantity in the period is at or above the 75th percentile of items sold in the period
FR-STK-3	Normal	Sold in the period, below the fast-moving threshold
FR-STK-4	Slow-moving	Last sale more than 90 days ago and within 180 days
FR-STK-5	Dead stock	No sale for 180 days or more, and current stock above zero
FR-STK-12	Never sold	No sale anywhere in the synced history, and current stock above zero
FR-STK-19	Not classified	No current stock and no sale in the period; excluded from movement views

ID	Requirement
FR-STK-6	Sales quantity is net of linked sales returns; cancelled vouchers are excluded
FR-STK-13	An item with no sale date at all is Never Sold, never Slow-moving
FR-STK-14	Never Sold is its own filterable category
FR-STK-7	Thresholds are stored in company_settings
FR-STK-20	Ranking by quantity compares items measured in different units. Whether fast-moving should instead be ranked by sales value is OPEN – BUSINESS DECISION (default: quantity, as specified; recommendation: value)

11.2 Units

ID	Requirement
FR-STK-8	A live export determines whether transaction units, base units and conversion factors are available (gate G27)
FR-STK-9	If available, all quantities are converted to each item’s base unit before use
FR-STK-10	If not, multi-unit conversion is out of scope for v1; items with more than one transaction unit are flagged and the limitation is stated on the stock view

11.3 Current stock quantity

ID	Requirement
FR-STK-15	Current stock quantity is the Tally-computed closing quantity on the sync date, stored in stock_snapshots, because inventory-only vouchers are not synced
FR-STK-16	Every stock classification shows the snapshot date it used
FR-STK-17	Stock is reported at company level; godown and batch breakdowns are out of scope for v1

12. Anomaly Detection (Optional)

12.1 Feature flag behaviour

Condition	Rules engine	Anomaly records	MCP / Claude calls	Anomalies view
FEATURE_ANOMALY_DETECTION off	Does not run	None created	None	Hidden
On, Claude reachable	Runs	Created with full evidence	Explanation requested	Evidence and explanation shown
On, Claude unreachable	Runs	Created with full evidence	Attempted, fails safely	Evidence shown; explanation marked unavailable

12.2 Flow

Database → deterministic rules engine (SQL/Python) → anomaly flags (numeric evidence) → MCP server tool get_anomaly_evidence(anomaly_id) → MCP client → Claude → plain-language explanation → stored with the flag → dashboard
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The application’s rules engine finds every anomaly and computes every number. Claude only reads the evidence for one anomaly and writes an explanation of it.

12.3 Rules

ID	Requirement
FR-3.1	Nothing in this section runs while the feature flag is off
FR-3.2	Unusually large transaction: for each party ledger, compute the mean and standard deviation of transaction amounts over a trailing window (default 180 days). Flag a transaction above mean + 3 standard deviations, or above the previous maximum × a configurable multiplier (default not set; rule inactive until configured, IMPLEMENTATION TEST)
FR-3.3	Possible duplicate: same party ledger, same amount, within a configurable window (default 3 days)
FR-3.4	Every flag stores the transaction amount, historical average, historical maximum, deviation percentage and the rule triggered
FR-3.5	The MCP tool returns only the stored evidence for one anomaly; it never returns bulk data and never runs arbitrary queries
FR-3.6	Claude is asked only to explain the supplied evidence; it is never asked to calculate a number
FR-3.7	Evidence and explanation are shown in separate, clearly labelled areas
FR-3.8	A user can mark a flag “reviewed” or “not an issue”; this is recorded in audit_logs
FR-3.9	Every MCP tool call is recorded in ai_tool_log

Example evidence: transaction ₹4,50,000; historical average ₹70,000; previous maximum ₹1,20,000; deviation +543%; rule: amount above mean + 3 SD.

12.4 Why Claude, and does the system need it?

Claude turns numeric evidence into a sentence a non-technical owner can act on. It is not needed for detection or for any figure. With the flag on and Claude unavailable, anomalies are still found and shown with their evidence. With the flag off, the core system is unaffected.

13. Dashboard, Drill-Down and Exports

13.1 Dashboard

ID	Requirement
FR-4.1	The home view shows sales, cash and bank position, receivables, payables, last sync time and status, and reconciliation status
FR-4.2	Sections: Sales, Purchases, Cash Flow, Balances, Aging, Payment Behaviour, Customers, Products, Expenses, Stock, Reconciliation, Data Quality, Anomalies (only if enabled), Settings (role-restricted)
FR-4.3	Filters: date range, customer, product, cost centre
FR-4.4	The Agents view lists every Agent with status, versions, last heartbeat, Tally uptime and queue status
FR-4.5	The Data Quality view lists unresolved groups, unresolved voucher types, missing opening balances, unsupported allocations, unlinked credit/debit notes, missing masters and MISSING_IN_TALLY vouchers

13.2 Drill-down

ID	Requirement
FR-DD-1	Revenue → customer → vouchers → voucher detail (ledger entries and line items)
FR-DD-2	Product revenue → product → contributing line items
FR-DD-3	Expense category → ledger → vouchers
FR-DD-4	Product Attribution Difference (or Unattributed/Non-product Revenue) → contributing vouchers
FR-DD-5	Every drilled list totals exactly to the figure it expands

13.3 Exports

ID	Requirement
EXP-1.1	Every CSV/PDF export contains the report title, company name, date range, active filters, generation timestamp (in company_timezone), summary figures and the detailed rows shown on screen
EXP-1.2	PDF exports include the charts shown on screen
EXP-1.3	Exports use the same query functions as the screen, so their figures match exactly
EXP-1.4	Sales exports show Total Sales Revenue, Product-attributed Revenue and whichever difference label the dashboard is showing, as separate lines
EXP-1.5	Mapped custom fields are included in detailed rows
EXP-1.6	The export endpoint re-checks the user's role and company access before producing a file

14. Authentication, Access Control and Security

14.1 Roles and permissions

Action	Owner	Accountant	Admin/Developer
View dashboard and financial data	✓	✓	✓ (companies assigned to them)
Run manual sync	✓	✓	✓
Create or change sync schedules	✓	✗	✓
Export reports	✓	✓	✓
View reconciliation and data-quality views	✓	✓	✓
View sync and system logs	✓	✗	✓
Review / dismiss anomalies	✓	✓	✗
Change feature flags and settings	✓	✗	✓
Register, rotate, revoke Agents; change Tally connection settings	✓	✗	✓
Manage custom field mappings	✓	✗	✓
Manage users and roles	✓	✗	✗

ID	Requirement
RBAC-1.1	Permissions are enforced on the server for every endpoint; hiding a

	button is never the only control
RBAC-1.2	A user's roles are held per company (user_roles)

14.2 Security requirements

ID	Requirement
SEC-1.1	Users authenticate with email and password (bcrypt hash); the backend issues JWTs that expire within 24 hours, with a refresh flow
SEC-1.2	Every endpoint checks role claims on the server
SEC-1.3	All traffic uses HTTPS with TLS 1.2 or later; plain HTTP is rejected
SEC-1.4	CORS allows only the known frontend origin(s)
SEC-1.5	CSRF protection applies to any cookie-based session flow; bearer tokens in headers are used by default
SEC-1.6	All database access uses parameterized queries through the ORM; no SQL is built by string concatenation
SEC-1.7	Every company-scoped query includes a company_id filter taken from the authenticated session, never from client input
SEC-1.8	Agent authentication follows Section 4.3–4.4
SEC-1.9	Rate limits: 100 requests/minute per IP unauthenticated; 1,000 requests/minute per authenticated user
SEC-1.10	Every request body is validated against a schema (Pydantic)
SEC-1.11	Exports re-check permission and company scope on the server
SEC-1.12	Only the evidence for a single anomaly is ever sent to the LLM; no bulk data, no unrelated narration text
SEC-1.13	audit_logs is append-only; the application has no update or delete path for it
SEC-1.14	Secrets (database credentials, JWT signing key, Claude API key) are held in environment configuration or a secret store, never in source code
SEC-1.15	The database user used by the application has no rights to alter the schema in production

15. Logging and Audit

Log	Contents	Storage	Retention
System log	Application errors, request traces	Log files / log service	30 days
Sync log	One row per sync run and per record-level failure	sync_runs, sync_errors	At least 90 days
Audit log	Who did what, when, to which company and data range, before/after values, success or failure	audit_logs	For as long as the company's data is kept
AI tool log	Every MCP tool call: tool, parameters, result summary, status	ai_tool_log	At least 90 days

ID	Requirement
LOG-1.1	Audited actions include: login, sync requests, schedule changes, Agent registration/rotation/revocation, setting and feature-flag changes, user and role changes, anomaly reviews, exports, and system changes to vouchers and masters (modifications, cancellations, missing and reappeared records)
LOG-1.2	Each audit entry records user (or “system”), action, company, entity, data range where relevant, before/after values where relevant, result, and timestamp

16. Failure Modes

Condition	Detection	Behaviour	What the user sees	Recovery
Tally closed	Agent cannot connect	Nothing pulled; no data changed	“Tally unreachable since ”	Resumes when Tally runs
Windows user logged off	Agent cannot connect	As above	“Tally not running — the	Log in and start Tally;

			Windows user may have logged off"	disconnect instead of logging off
Tally XML server disabled	Connection refused	TALLY_SERVER_DISABLED	Message naming the Tally setting to enable	Enable the server
Required TDL not loaded	Unknown report response	TDL_NOT_LOADED; no fallback	Configuration error	Load the TDL files
Named company not loaded	Targeted request fails	COMPANY_NOT_LOADED; nothing pulled	Warning on the Agent	Load the company, or update its stored name
Company mismatch	GUID check fails	COMPANY_MISMATCH; sync halted	Warning on the Agent	Re-register
Tally export stalls	No response within timeout	Retry at half batch, then fail the segment	Run PARTIAL, TALLY_EXPORT_TIMEOUT	Lower the batch size
Long Tally uptime	Uptime above threshold	Sync continues; advisory raised	"Restart recommended"	Restart Tally, run reconciliation
Malformed XML	Parser error	Record logged in sync_errors; batch continues	Run PARTIAL with error count	Investigate; fix TDL if needed
Debit/credit inconsistency	A voucher's normalized entries do not balance	Voucher not written; flagged	Listed in Data Quality	Investigate parser mapping
Agent loses internet	Upload fails	Local queue, backoff retries	"Last contact "	Automatic on reconnect
Agent queue full	Queue limit reached	Pulling stops; uploads keep retrying	QUEUE_FULL	Restore connectivity
Agent crashes	Heartbeats stop	Agent OFFLINE; running command becomes FAILED_AGENT_LOST	Agent offline; command failed	Restart Agent; issue a new command
Selected Agent offline	No recent heartbeat	Command stays PENDING	"Waiting — Agent offline since "	Claimed on return, or expires
Several Agents, no Agent chosen	Request lacks agent_id	Request rejected	Agent selector shown	Choose an Agent
Two Agents sync the same collection	Lease already held	Second gets SYNC_LOCKED	"Sync in progress by "	Retries after release
Duplicate command claim	Compare-and-set fails	Second claim rejected	None	None needed
Stale record	Lower ALTERID	Rejected, STALE_ALTERID	Sync log entry	None needed
Partial batch failure	Transaction failure	Batch rolled back; watermark not advanced	Run PARTIAL	Next run resumes
Invalid or revoked credential	Auth fails	401 CREDENTIAL_INVALID / AGENT_REVOKED	Agent-side error	Re-provision or re-register
Incompatible Agent or TDL version	Version below minimum	INCOMPATIBLE; no commands	"Agent update required"	Update Agent or TDL
Unresolved group / voucher type	Resolution fails	Excluded from classified metrics	Data Quality list	Fix in Tally
Missing opening balance	No record for the year	Balance shown as unavailable	"Opening balance unavailable"	Re-sync masters
Unsupported bill allocation	Unknown type	Excluded from aging	Review list	Classify; update parser if needed
Unlinked credit/debit note	No bill link	Not subtracted	Unclassified Adjustments	Manual review
Master missing in Tally	Absent from key list or reconciliation	MISSING_IN_TALLY; history kept	Missing masters list	Automatic if it returns
Reconciliation with zero Tally value	Tally value is 0	Exact-match rule (Section 9.2)	PASS/FAIL	None needed
Mapped custom field missing	Absent from export	Stored as null, UDF_NOT_FOUND	Sync log	Fix mapping or customisation
Invalid time zone or financial-year start	Settings validation	Save rejected; quarter views show "financial year not configured"	Validation message	Correct the setting
ALTERID gate row failed	Validation table	Collection uses full-pull-only sync	"Full sync only" indicator	Fix and re-validate
Backend unavailable	Agent calls fail	Agent queues	Service-unavailable page	Operations recovery
Database unavailable	Backend DB calls fail	HTTP 503; no partial writes	Service-unavailable page	Operations recovery
MCP server or Claude unavailable	Tool call fails	Evidence still created and shown	"Explanation unavailable"	Automatic when restored

17. Non-Functional Requirements

17.1 Performance targets

ID	Metric	Target
PERF-1.1	Dashboard summary load	≤ 3 seconds
PERF-1.2	Full sync of the benchmark dataset	≤ 30 minutes
PERF-1.3	Incremental sync (about 500 changed records)	≤ 2 minutes
PERF-1.4	Anomaly evidence tool call	≤ 2 seconds

17.2 Benchmark conditions

Parameter	Value
Dataset	100,000 vouchers, 500,000 voucher entries, 5,000 ledgers, 10,000 stock items; seeded, reproducible, version-controlled
Concurrent users	10
Backend	4 vCPU, 8 GB RAM
PostgreSQL	Managed instance, 4 vCPU, 16 GB RAM, SSD
Agent machine	2 vCPU, 4 GB RAM (also running TallyPrime)
Network	≥ 10 Mbps between Agent and backend

ID	Requirement
PERF-VAL-1	A benchmark result is valid evidence only if it records: Windows edition and build, Tally machine CPU model and RAM, storage type and free space, TallyPrime version and build, dataset identifier, Agent version and configuration, and measured bandwidth and latency during the run
PERF-VAL-2	These are benchmark conditions for repeatable testing, not a guarantee for every customer environment

17.3 Reliability and scalability

ID	Requirement
NFR-REL-1	The core system works fully with anomaly detection disabled or the AI service unavailable
NFR-REL-2	No failure leaves a voucher partly written
NFR-SCALE-1	The schema supports multiple companies without redesign

17.4 Usability

ID	Requirement
NFR-UI-1	Every view is usable at 360 px width or wider without page-level horizontal scrolling; wide tables scroll inside their own container
NFR-UI-2	Charts, filters and drill-downs work by touch
NFR-UI-3	A non-technical owner can read the dashboard without training

17.5 Dates, time zones and quarters

ID	Requirement
TZ-1.1	All date grouping, aging “today”, report periods, date-based schedules and displayed timestamps use <code>company_timezone</code> , never the server time zone
TZ-1.2	Day boundaries are computed in <code>company_timezone</code> , so a transaction at 11:58 PM local time belongs to that local day
Q-1.1	Quarters are financial quarters counted from <code>financial_year_start</code> , unless a company chooses calendar quarters
Q-1.2	With a 1 April start: Q1 Apr-Jun, Q2 Jul-Sep, Q3 Oct-Dec, Q4 Jan-Mar. A 1 January start gives calendar quarters by the same rule

17.6 Maintainability

ID	Requirement
NFR-MAINT-1	TDL definitions, field mappings, parser and MCP tool definitions are kept in separate, version-controlled modules

NFR-MAINT-2	Accounting rule queries are defined once and reused by dashboard, drill-down and export
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18. Configuration

18.1 Feature flags (feature_config, boolean only)

Flag	Default
FEATURE_ANOMALY_DETECTION	Off

18.2 Company settings (company_settings)

Key	Default
classification.sales_groups	Sales Accounts
classification.purchase_groups	Purchase Accounts
classification.expense_groups	Direct Expenses, Indirect Expenses
classification.cash_bank_groups	Cash-in-Hand, Bank Accounts
classification.tax_groups	Duties & Taxes
analytics.taxable_value_mode	true
analytics.top_n_default	10
analytics.quarter_mode	financial
cashflow.include_journal	false
aging.bucket_boundaries	[30, 60, 90]
payment.window_days / payment.min_settlements	365 / 3
stock.measurement_period_days	90
stock.fast_percentile	75
stock.slow_threshold_days / stock.dead_stock_days	90 / 180
reconciliation.money_absolute_tolerance / money_percentage_tolerance	₹1.00 / 0.01%
reconciliation.quantity_absolute_tolerance / quantity_percentage_tolerance	0 / 0.5%
sync.incremental_interval	Hourly
sync.full_reconciliation_interval	Daily
sync.key_list_interval	Every incremental run
sync.db_commit_batch	500
agent.poll_interval_seconds	30
agent.offline_threshold_minutes	5
agent.command_claim_timeout_minutes	10
agent.tally_uptime_advisory_days	7
anomaly.history_window_days	180
anomaly.deviation_sd	3
anomaly.max_multiplier	Not set (rule inactive)
anomaly.duplicate_window_days	3

Agent-level settings (host, port, company name, extraction batch size, queue limits) are stored on the agent's row.

19. External Interfaces and API

19.1 Tally interface

The Agent sends HTTP POST requests containing XML export requests that reference the project's TDL Reports to `http://<tally-host>:<port>` (default `localhost:9000`), naming the registered company, and parses the XML responses.

19.2 Backend API

All endpoints use HTTPS. User endpoints require a JWT; Agent endpoints require an Agent credential. Every company-scoped endpoint enforces `company_id` from the session and verifies that any `agent_id` belongs to that company.

Area	Method and path	Access
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Login	POST /auth/login	Public
Token refresh	POST /auth/refresh	JWT
Registration token	POST /companies/{id}/agents/register-token	Owner/Admin
Agent registration	POST /agent/register	Registration token
Agent heartbeat and polling	POST /agent/heartbeat	Agent credential
Command claim / progress / result, batch upload	POST /agent/commands/{id}/claim, /progress, /result, /batches	Agent credential
List Agents	GET /companies/{id}/agents	JWT
Rotate / revoke Agent	POST /companies/{id}/agents/{agent_id}/rotate-credential, /revoke	Owner/Admin
Agent Tally settings	PUT /companies/{id}/agents/{agent_id}/tally-settings	Owner/Admin
Manual sync	POST /companies/{id}/agents/{agent_id}/sync	Owner/Accountant/Admin
Schedules	GET/POST/PUT /companies/{id}/sync-schedules	Owner/Admin
Sync status, errors, lease status	GET /companies/{id}/sync/status, /errors, /lease-status	JWT (errors: Owner/Admin)
Reconciliation	GET /companies/{id}/reconciliation; POST .../reconciliation/run	JWT
Analytics	GET /companies/{id}/analytics/{metric} (sales, purchases, cash-flow, balances, aging, payment-behaviour, customers, products, expenses, stock) with from, to, filters, top_n	JWT
Drill-down	GET /companies/{id}/analytics/{metric}/drilldown	JWT
Voucher detail	GET /companies/{id}/vouchers/{voucher_id}	JWT
Masters	GET /companies/{id}/masters/groups, /voucher-types	JWT
Data quality	GET /companies/{id}/data-quality	JWT
Exports	GET /companies/{id}/exports/{report}?format=csv pdf	JWT
Settings and flags	GET/PUT /companies/{id}/settings	Owner/Admin to change
Custom field mappings	GET/PUT /companies/{id}/settings/custom-fields	Owner/Admin
Anomalies	GET /companies/{id}/anomalies; POST .../anomalies/{id}/review	JWT; review Owner/Accountant
Users	GET/POST/PUT /companies/{id}/users	Owner

Errors return a JSON body with a machine-readable code (for example SYNC_LOCKED, COMPANY_MISMATCH, CREDENTIAL_INVALID) and a message; validation errors use 422, authorization errors 403, authentication errors 401.

19.3 MCP interface (optional)

The MCP server exposes one read-only tool, `get_anomaly_evidence(anomaly_id)`, returning the stored evidence for that anomaly. It is available only when anomaly detection is enabled.

20. Use Cases

UC-1 Onboarding. Owner creates the company, generates a registration token, installs and names the Agent, enters the token; the Agent confirms the Tally company; the Owner triggers the initial full sync.

UC-2 Scheduled incremental sync. A schedule creates a command for its Agent; the Agent claims it, acquires the leases, pulls changes above each watermark, runs the key-list check, uploads batches, and completes; the dashboard updates.

UC-3 Manual sync with several Agents. The Owner clicks “Sync Now”, chooses an Agent, and only that Agent claims the command.

UC-4 Modified voucher. A voucher edited in Tally is pulled with a higher ALTERID; its children are replaced in one transaction and the change is audited.

UC-5 Cancelled and deleted vouchers. A cancelled voucher becomes CANCELLED; a deleted voucher becomes MISSING_IN_TALLY after the key-list check; neither row is deleted.

UC-6 Stale record. A retried older batch arrives after a newer one; the older record is rejected and logged.

UC-7 Concurrent Agents. Two Agents try to sync the same collection; one gets the lease, the other receives SYNC_LOCKED.

UC-8 Agent crash. An Agent crashes mid-sync; the command becomes FAILED_AGENT_LOST; the Owner issues a new command.

UC-9 Credential rotation. The Owner rotates an Agent’s credential, sees the new value once, and enters it in the Agent.

UC-10 Reconciliation review. The Accountant opens Reconciliation, sees Tally and local values with differences, and investigates any FAIL.

UC-11 Analytics review and export. The Owner reviews sales, cash position and aging for a period, drills into a customer’s vouchers, and exports the view.

UC-12 Product Attribution Difference review. The Accountant drills into the difference to see which vouchers contain revenue not tied to stock items.

UC-13 Anomaly review (optional). A large transaction is flagged with evidence and, if available, an explanation; the Accountant marks it reviewed.

UC-14 Data-quality review. The Admin checks unresolved groups, unresolved voucher types, missing opening balances and unlinked notes.

21. Technology Stack

Layer	Choice	Reason
Tally customisation	TDL Collections and Reports	Controlled, stable export shape
Agent	Python, packaged as a Windows Service (NSSM or pywin32 wrapper, IMPLEMENTATION TEST)	Shares parsing code with the backend; Windows matches Tally’s platform
Tally connectivity	requests + lxml	HTTP XML requests and tolerant XML parsing
Agent local queue	SQLite	Simple embedded store
Backend	Python, FastAPI, Pydantic	Async API, strict request validation
Database	PostgreSQL	Relational integrity, concurrent access, JSONB for custom fields
ORM / migrations	SQLAlchemy, Alembic	Parameterized queries; versioned schema
Scheduling	APScheduler (backend schedules and reconciliation)	Lightweight
Analytics	SQL views and pandas	Deterministic aggregation
Frontend	React, Tailwind CSS, Recharts	Routing for drill-down, responsive layout, charts
Authentication	JWT, bcrypt (passlib)	Standard, well understood
Optional anomaly explanation	MCP server (Python mcp SDK) + Claude	Explanation only
Testing	pytest; Playwright for end-to-end	Backend/Agent and UI coverage
Deployment	Docker (backend), Windows installer (Agent)	Matches the topology
Version control	Git	Includes TDL sources

22. Backup and Recovery

ID	Requirement
BKP-1.1	PostgreSQL: daily full backup plus continuous WAL archiving, retained at least 30 days
BKP-1.2	A restore is tested before go-live: restore to staging and compare row counts with a known reconciliation snapshot
BKP-1.3	A failed backup raises an alert; it is never skipped silently
BKP-1.4	Recovery targets: RPO ≤ 24 hours, RTO ≤ 4 hours (OPEN – BUSINESS DECISION to confirm)
BKP-1.5	A fresh full sync recovers records that still exist in Tally. Records deleted from Tally can be recovered only from a database backup
BKP-1.6	Agent local queues are transient and not backed up; data in them is re-pulled from Tally if lost

23. Testing Strategy

ID	Requirement

TEST-1.1	TDL/XML contract tests: fixed XML fixtures run through the parser to a normalized model and compared with expected database rows
TEST-1.2	Fixtures cover: new, modified and cancelled vouchers; multiple ledger entries; inventory entries; bill allocations of every type; cost-centre allocations; missing optional fields; malformed XML; unexpected structure; a three-level group chain; a group with a missing parent; a custom voucher type derived from Sales; an unresolvable voucher type; opening balances; stock snapshots; debit and credit entries on sales, purchase, receipt, payment and tax ledgers
TEST-1.3	Contract tests run in CI without a live Tally connection
TEST-1.4	Malformed fixtures must produce logged errors, never unhandled exceptions
TEST-2.1	Accounting unit tests for every ACC rule, reconciliation example (Section 9.3), aging bucket and stock class
TEST-2.2	The validation gate (Section 7) is executed on a live TallyPrime installation and its evidence recorded
TEST-3.1	Concurrency: two simulated Agents contend for one lease; exactly one succeeds
TEST-3.2	Recovery: a simulated Agent stops mid-command; the command becomes FAILED_AGENT_LOST and is not reassigned
TEST-3.3	Fault injection: processes killed mid-voucher and mid-batch leave no partial vouchers
TEST-4.1	Before implementation, tally-database-loader is run against the project's Tally test company and its output kept as comparison evidence for gate items
TEST-4.2	Balance test: computed ledger balances match Tally's closing balances for every ledger in the test dataset
TEST-5.1	End-to-end tests cover RBAC (403 on forbidden actions), drill-down totals, export/screen consistency and 360 px layouts
TEST-5.2	Performance tests run under Section 17.2 conditions with full recording

24. Acceptance Criteria

Each criterion is written as Given / When / Then and is the basis of a test case with the same ID.

24.1 Synchronization and identity

AC-01 Full sync. Given Tally holds X active vouchers, when a full sync runs, then the database holds exactly X corresponding vouchers keyed by (company_id, GUID), and running it again adds no rows.

AC-02 Modified voucher. Given voucher 100 was synced at ₹10,000, when it is edited in Tally to ₹15,000 and an incremental sync runs, then the local voucher shows ₹15,000 and audit_logs records the old and new values.

AC-03 Cancelled voucher. Given a synced voucher is cancelled in Tally, when an incremental sync runs, then its status is CANCELLED, the row still exists, and it is excluded from standard analytics.

AC-04 Deleted voucher. Given a synced voucher is deleted in Tally, when the next incremental sync and key-list check complete, then its status is MISSING_IN_TALLY, the row still exists, and it appears in the Data Quality view.

AC-05 Partial failure. Given a sync fails after 50,000 of 100,000 records, when the next sync runs, then it resumes from the last committed watermark, no duplicates exist, and the failed run is recorded as PARTIAL.

AC-06 Stale record. Given a local voucher with ALTERID 108, when a record for it with ALTERID 106 arrives, then the local voucher is unchanged and STALE_ALTERID is logged.

AC-07 Independent watermarks. Given the Voucher watermark is 500 and the Ledger watermark is 50, when an incremental sync runs, then each collection is pulled from its own watermark and a ledger change is neither skipped nor affected by the voucher watermark.

AC-08 Child replacement. Given a synced voucher is modified and no stable line identifier exists, when it is synced, then all its child rows are replaced in one transaction and no orphaned or duplicate child rows remain.

AC-09 Sync lease. Given two Agents request the same company and collection at the same time, when both try to acquire the lease, then exactly one succeeds and the other receives SYNC_LOCKED without pulling data.

AC-10 Master disappears. Given a synced ledger is missing from a full reconciliation, when reconciliation completes, then the ledger is MISSING_IN_TALLY, is not deleted, and historical vouchers still reference it.

AC-11 Master reappears. Given a MISSING_IN_TALLY ledger appears again with the same GUID, when it is synced, then it becomes ACTIVE and the event is audited.

AC-12 Failed ALTERID gate. Given a validation-gate row for a collection is FAILED, when syncs run, then that collection is synced by full pull only and the dashboard shows "Full sync only".

24.2 Agent

AC-13 Registration. Given a valid registration token, when the Agent registers, then an Agent credential is issued and bound to exactly one company.

AC-14 Second Agent, different company. Given a company already bound to Tally GUID G1, when a new Agent registers reporting GUID G2, then registration is rejected with `COMPANY_MISMATCH`.

AC-15 Revocation. Given an Agent is revoked, when it next sends a heartbeat, then it receives `AGENT_REVOKED` and no commands.

AC-16 Rotation. Given an Agent's credential is rotated, when the Agent uses the old credential, then it receives `401 CREDENTIAL_INVALID`, and the new credential works.

AC-17 Manual sync by polling. Given the Owner clicks "Sync Now", when the Agent next polls, then it claims the command, which moves through `CLAIMED`, `RUNNING` and `COMPLETED`, each shown on the dashboard.

AC-18 Routing. Given two active Agents, when a sync request has no `agent_id`, then it is rejected; when it names one Agent, then only that Agent receives the command.

AC-19 Offline Agent. Given the chosen Agent is offline, when a command is created for it, then it stays `PENDING` with "Agent offline" shown, and is claimed when the Agent returns unless it expires.

AC-20 Agent crash. Given a command is `RUNNING` and the Agent stops heartbeating past the lease deadline, then the command becomes `FAILED_AGENT_LOST` and is not reassigned.

AC-21 Queue full. Given the Agent's queue reaches its limit, then the Agent stops pulling new data, keeps retrying uploads and reports `QUEUE_FULL`.

AC-22 Version check. Given an Agent below the minimum version, when it sends a heartbeat, then it is marked `INCOMPATIBLE` and receives no commands.

AC-23 Company targeting. Given two companies are loaded in Tally and the other one is active, when the Agent syncs, then it pulls only from its registered company; if that company is closed, it reports `COMPANY_NOT_LOADED` and pulls nothing.

AC-24 Batch limit. Given a configured batch size of 15,000, then the setting is rejected; given a Tally request exceeds the timeout, then it is retried once at half size and logged as `TALLY_EXPORT_TIMEOUT`.

AC-25 Uptime advisory. Given Tally's uptime exceeds the threshold, then a restart advisory is shown against that Agent.

24.3 Accounting and analytics

AC-26 Sales is single-sided. Given a Sales voucher debiting a customer ₹10,000 and crediting Sales ₹10,000, when Total Sales Revenue is computed, then it is ₹10,000 — not ₹20,000 and not ₹0.

AC-27 Purchases. Given a Purchase voucher with matching debit and credit entries, then Purchase Value counts only the debit on the purchase ledger.

AC-28 Nested groups. Given a ledger under "Sales – Online", itself under Sales Accounts, then its qualifying credits are included in Total Sales Revenue.

AC-29 Custom voucher type. Given a "POS Invoice" voucher type derived from Sales, then its vouchers are included in sales exactly like Sales vouchers.

AC-30 Customer attribution. Given a Sales voucher with exactly one customer ledger and ₹50,000 revenue, then ₹50,000 is attributed to that customer.

AC-31 Cash sale. Given a Sales voucher with no customer ledger, then its revenue is Unattributed Customer Revenue, and customer-attributed plus unattributed revenue equals Total Sales Revenue.

AC-32 Product Attribution Difference. Given a Sales voucher with ₹92,000 of stock-item lines and an ₹8,000 service line, and gate G28 not yet passed, then Total Sales Revenue is ₹100,000, Product-attributed Revenue ₹92,000, and ₹8,000 is shown as Product Attribution Difference, not as service revenue.

AC-33 Top-N. Given $N = 10$ and 25 customers with sales, then the Top view shows 10, does not claim their sum equals total sales, and "View All" shows 25.

AC-34 Sign normalization. Given raw Tally entries, when parsed, then `accounting_direction`, `is_debit` and `amount_absolute` are correct, and no analytics query reads `amount_raw`.

AC-35 Linked return. Given a Credit Note linked by bill reference to a Sales invoice, then it reduces Total Sales Revenue.

AC-36 Unlinked note. Given a Credit Note with no bill link, then it is `UNLINKED_CREDIT_NOTE`, is not subtracted, and appears in Unclassified Adjustments.

AC-37 Cash flow. Given a Receipt of ₹5,000, a Payment of ₹3,000 and a Contra transfer of ₹10,000, then the net cash flow shows +₹5,000 and –₹3,000, and the Contra is excluded.

AC-38 Ledger balance. Given a bank ledger with a ₹50,000 debit opening balance and ₹20,000 net debit movements up to date D, then its balance on D is ₹70,000 debit and matches Tally within tolerance.

AC-39 Screen and export match. Given the same filters, then dashboard, drill-down total and CSV/PDF export show identical figures.

24.4 Reconciliation

AC-40 Display. Given Tally total X and local total Y, then reconciliation shows X, Y, the absolute and percentage difference, and PASS or FAIL.

AC-41 Both tolerances exceeded. Given Tally ₹100,000 and local ₹100,012 with tolerances ₹1 and 0.01%, then the result is FAIL.

AC-42 One tolerance met. Given Tally ₹100,000 and local ₹100,010, then the result is PASS.

AC-43 Zero Tally value. Given a Tally value of 0, then a local value of 0 passes and a local value of 50 fails.

AC-44 Quantity tolerance. Given Tally stock of 100 units and local 98 with tolerances 0 units and 0.5%, then the result is FAIL.

24.5 Aging, payment behaviour and stock

AC-45 Overdue bucket. Given a bill with ₹20,000 outstanding and a due date 45 days ago, then it appears in 31–60 days with ₹20,000.

AC-46 Not yet due. Given a bill due in 5 days, then it is “Not yet due”, never overdue and never negative.

AC-47 No due date. Given a bill without a due date, then it is “Due date unavailable”.

AC-48 Advance. Given a customer receipt allocated as Advance, then it appears under customer “Unadjusted Advances” and in no aging bucket.

AC-49 On Account. Given a payment allocated On Account, then it appears under “On-Account / Unallocated” and in no aging bucket.

AC-50 Receivable and payable separation. Given a customer advance and a supplier advance of the same amount, then they appear on their respective sides and are not netted.

AC-51 Payment behaviour. Given a ₹10,000 bill dated 1 June, settled ₹6,000 on 11 June and ₹4,000 on 21 June, then average days to pay is 14.

AC-52 Never sold. Given an item with stock above zero and no sales history, then it is Never Sold, not Slow-moving or Dead.

AC-53 Dead stock from snapshot. Given an item with a snapshot quantity of 40 and no sale for 200 days, then it is Dead stock and the snapshot date is shown.

AC-54 Unit limitation. Given unit data is unavailable and an item sold in two units, then the item is flagged and the stock view states the limitation.

24.6 Anomaly detection

AC-55 Flag off. Given anomaly detection is off, when syncs complete, then no anomaly records are created and no MCP or Claude calls are made.

AC-56 Large transaction. Given a transaction above mean + 3 standard deviations for its party, then an anomaly record is created with amount, average, maximum, deviation and rule.

AC-57 Duplicate. Given two vouchers for the same party and amount within 3 days, then a duplicate anomaly is created referencing both.

AC-58 Claude unavailable. Given anomaly detection is on and Claude is unreachable, then anomalies are still created and shown with evidence, and the explanation is marked unavailable.

24.7 Security, configuration and non-functional

AC-59 RBAC. Given an Accountant, when they call a Settings or user-management endpoint, then the API returns 403 regardless of the UI.

AC-60 Tenant isolation. Given a user of company A, when they request any company B resource, then they receive 403 and no data.

AC-61 Custom field. Given a mapped custom voucher field, when a voucher with it is synced, then the value appears in drill-down and export and no analytics total changes.

AC-62 Time zone. Given company_timezone Asia/Kolkata and a transaction at 11:58 PM IST on day 1, then daily grouping places it on day 1 regardless of server time zone.

AC-63 Financial quarter. Given a 1 April financial year start, then an August transaction falls in Q2.

AC-64 Performance. Given the Section 17.2 environment and dataset, when the performance suite runs, then the Section 17.1 targets are met and every PERF-VAL-1 field is recorded.

AC-65 Responsive layout. Given a 360 px viewport, then no view scrolls horizontally at page level and filters and drill-downs work by touch.

AC-66 Contract test. Given the malformed-XML fixture, when parsed, then an error is logged and no unhandled exception occurs.

25. Requirement Traceability

Requirement family	Objective	Component	Tables	Interface	UI	Acceptance criteria
FR-1.x TDL exports	Controlled extraction	TDL, Agent	—	Tally XML	—	AC-01, AC-66
SYNC-1.x watermarks	Efficient incremental sync	Sync engine	sync_watermarks	Agent batch upload	Sync status	AC-07
SYNC-3.x stale protection	No rollback of data	Sync engine	masters, vouchers	Agent batch upload	Sync log	AC-06
SYNC-4.x lease	No concurrent writers	Sync engine	sync_watermarks	Lease status API	Sync status	AC-09
SYNC-5.x deletion detection	Timely removal from analytics	Agent, sync engine	vouchers, masters	Key-only TDL export	Data Quality	AC-04
SYNC-6.x batching, partial failure	Integrity	Sync engine	sync_runs, sync_errors	Agent batch upload	Sync status	AC-05
DR-4.x identity	No duplicates	Database	all synced tables	—	—	AC-01
DR-VE-x child replacement	Integrity on modification	Sync engine	voucher child tables	—	Voucher detail	AC-02, AC-08
DR-ML-x master lifecycle	Keep history	Sync engine	masters	Reconciliation	Missing masters	AC-10, AC-11
DR-UDF-x custom fields	Customisation support	Sync engine	custom_field_mappings	Settings API	Settings, drill-down	AC-61
VAL-1.x gate	Verified Tally behaviour	Project process	—	Live Tally	Sync mode indicator	AC-12
AGT registration, auth, rotation	Secure onboarding	Agent, backend	agents, agent_registration_tokens	Agent APIs	Agents view	AC-13-AC-16
AGT-1.x commands	Manual sync, recovery	Agent, backend	agent_commands	Heartbeat, command APIs	Sync status	AC-17, AC-20
RTE-1.x routing	Correct Agent	Backend	agent_commands, sync_schedules	Sync API	Agent selector	AC-18, AC-19
AGT-2.x queue	No data loss offline	Agent	local queue	—	Agents view	AC-21
VER-1.x versions	Compatibility	Agent, backend	agents	Heartbeat	Agents view	AC-22
AGT-5.x targeting	Correct company	Agent	agents	Tally XML	Agents view	AC-14, AC-23
AGT-4.x batch size	No stalled exports	Agent	agents	Tally XML	Sync log	AC-24
AGT-6.x runtime	Operational stability	Agent	agents	Heartbeat	Agents view	AC-25
ACC-DATA-x normalization	Sign-safe analytics	Parser	voucher_entries	—	—	AC-34
ACC-7.x / ACC-8.x classification	Correct metric membership	Sync engine, analytics	groups, voucher_types	Masters API	Data Quality	AC-28, AC-29
ACC-1.x revenue, purchases	Correct figures	Analytics	voucher_entries, ledgers	Analytics API	Sales, Purchases	AC-26, AC-27
ACC-6.x customer attribution	Honest attribution	Analytics	voucher_entries	Analytics API	Customers	AC-30, AC-31
ACC-1.8-1.10 product attribution	Honest attribution	Analytics	voucher_items	Analytics API	Products	AC-32
ACC-5.x returns	Correct net sales	Analytics	bill_allocations	Analytics API	Unclassified Adjustments	AC-35, AC-36
ACC-3.x cash flow	Correct cash view	Analytics	voucher_entries	Analytics API	Cash Flow	AC-37
ACC-9.x balances	Correct balances	Analytics	ledger_opening_balances	Balances API	Balances	AC-38
TOPN-1.x	Clear rankings	Analytics, frontend	company_settings	Analytics API	Top views	AC-33
ACC-4.x, EXP-1.x	Trustworthy	Analytics, export	—	Export API	All views	AC-39

consistency	exports					
REC-1.x reconciliation	Proof of match	Reconciliation job	reconciliation_results	TDL Totals	Reconciliation	AC-40-AC-44
FR-AGE, AGE-BILL	Correct aging	Analytics	bill_allocations	Analytics API	Aging	AC-45-AC-50
FR-PAY-x	Customer credit insight	Analytics	bill_allocations	Analytics API	Payment Behaviour	AC-51
FR-STK-x	Stock insight	Analytics	stock_snapshots, voucher_items	Analytics API	Stock	AC-52-AC-54
FR-3.x anomaly	Optional insight	Rules engine, MCP	anomaly_flags, ai_tool_log	MCP tool	Anomalies	AC-55-AC-58
RBAC, SEC-1.x	Protection of financial data	Backend	users, user_roles	All APIs	Settings	AC-59, AC-60
TZ, Q	Correct dates	Analytics	companies	Analytics API	All date views	AC-62, AC-63
PERF	Responsiveness	Whole system	—	—	—	AC-64
NFR-UI	Mobile use	Frontend	—	—	All views	AC-65

A complete matrix mapping every individual requirement ID to its test case is a required project deliverable before final acceptance.

26. Technical Reviewer Questions

Architecture and connectivity

1. *How does a cloud website reach Tally on a customer's LAN?* It doesn't. A local Agent makes outbound HTTPS calls to the backend, so nothing needs to reach into the customer network.
2. *Why not open a firewall port to Tally?* It depends on customer network setup and would expose an unauthenticated XML server to the internet.
3. *How does an outbound-only Agent receive "Sync Now"?* It polls every 30 seconds and collects any pending command addressed to it.
4. *Can one company have several Agents?* Yes, one per Tally installation. Each Agent serves exactly one Tally company.
5. *How is the Agent for "Sync Now" chosen?* Automatically if there is one active Agent; otherwise the user must choose.
6. *What happens if the Agent is offline?* The command waits as PENDING with a visible status, and is collected when the Agent returns or expires.
7. *Does a standby Agent take over automatically?* No. Failover is a manual action in this version.
8. *What if the Agent crashes mid-sync?* The running command's lease expires and it becomes FAILED_AGENT_LOST; a new command must be issued.
9. *What if several companies are open in Tally?* The Agent names its company in every request and confirms the result by GUID.
10. *What stops an Agent from being moved to another company?* The GUID check on every sync; a mismatch halts the Agent and requires re-registration.
11. *Can the Agent keep Tally running?* No. Tally must run in a logged-in Windows session; the Agent reports when it cannot reach Tally.

Tally, TDL and XML

12. *Why custom TDL instead of Tally's default reports?* To control exactly which fields are returned and keep the format stable.
13. *What if the TDL isn't loaded?* The Agent reports TDL_NOT_LOADED and does not fall back to default reports.
14. *How do you know Tally behaves as assumed?* A 29-item validation gate must pass on a live installation before dependent features are declared complete.
15. *Why limit the Tally request size?* Very large exports can stall Tally; requests default to 5,000 vouchers, maximum 10,000.
16. *How do you verify the XML contract?* Fixed XML fixtures are parsed in CI and compared with expected database rows.

Database and identity

17. *Why not use voucher number as the key?* Numbers repeat across financial years and can be edited; Tally's GUID is stable.
18. *How is a journal with several debits and credits stored?* As several voucher_entries rows under one voucher header.
19. *Why is there company_id everywhere?* So more companies can be added without changing the schema, and so every query can be scoped to one company.
20. *Why no closing balance column?* A balance depends on the date. Opening balances are stored and balances computed, then checked against Tally.

Synchronization

21. *How do you detect a modified voucher?* Its ALTERID is higher than the stored value.
22. *Why are watermarks per collection?* Collections change independently; one shared counter could skip changes.
23. *What if an older record arrives after a newer one?* It is rejected as stale; the newer data is kept.
24. *Why both a lease and stale protection?* The lease stops two Agents writing at once; stale protection stops out-of-order data from any writer.
25. *How do you update a modified voucher's lines without a line ID?* Its lines are deleted and re-inserted in one transaction.
26. *How are deletions detected?* A key-only list after each incremental sync, with the daily reconciliation as a backstop.
27. *What happens to a cancelled or deleted voucher?* It is kept with status CANCELLED or MISSING_IN_TALLY and excluded from standard analytics.

28. *How does a failed sync recover?* Committed batches stay, the watermark marks progress, and the next run resumes without duplicates.

Accounting correctness

29. *Why not sum every entry of a Sales voucher?* Debits and credits cancel out; only credits on sales ledgers count.
30. *What exactly counts as a sale?* Credits on ledgers whose primary group is Sales Accounts, on active vouchers whose base type is Sales, minus linked returns, excluding tax.
31. *Why use primary groups?* Groups are nested; checking only the immediate parent would miss sub-groups.
32. *What about custom voucher types?* Each is resolved to its base type, so a Sales-derived type counts as sales.
33. *Is every Credit Note a sales return?* No. Only one linked to an original sales bill; others are shown separately.
34. *Why can't product revenue equal total sales?* Invoices can include non-stock lines, and product-line amounts may be computed on a different basis; the gap is shown as Product Attribution Difference until proven.
35. *Why don't the Top 10 customers add up to total sales?* Top-N is a subset. The full customer list plus unattributed revenue does.
36. *What happens to a cash sale?* It is Unattributed Customer Revenue, never assigned to an invented customer.
37. *Why are Contra vouchers excluded from cash flow?* They move money between the company's own accounts.
38. *Are journal entries in cash flow?* Excluded by default; a company can enable them.
39. *Is GST analysed?* Not in v1 by default; revenue is at taxable value. Bringing GST in scope is a business decision.

Reconciliation, aging and stock

40. *How do you prove the database matches Tally?* Totals from a separate TDL Totals Collection are compared with local figures under stated tolerances.
41. *When does reconciliation pass?* When either the absolute or the percentage difference is within tolerance; it fails only when both are exceeded.
42. *What if Tally's value is zero?* It passes only if the local value is also zero.
43. *How is aging computed?* Per bill reference from due dates, with separate "Not yet due" and "Due date unavailable" categories.
44. *How do Advance and On Account affect aging?* They are not in buckets; they appear as separate lines, per customer or supplier side.
45. *What is Never Sold stock?* Stock above zero with no sale in the entire synced history.
46. *How do you know current stock without syncing delivery notes?* From Tally's own closing-quantity snapshot on each sync date.

Security, AI and operations

47. *How is data isolated between companies?* Every query is filtered by the company from the authenticated session, enforced on the server.
48. *How is the Agent authenticated without HMAC?* A per-Agent bearer credential over TLS, stored hashed and rotatable.
49. *Could Claude report a wrong number?* No. All numbers are computed first; Claude only explains them.
50. *Does the system work without Claude?* Yes. Anomalies are still detected and shown; only the explanation is missing.
51. *What data does Claude see?* Only the stored evidence for one anomaly.
52. *Can a fresh sync restore everything?* Only records still in Tally; deleted history needs a database backup.
53. *What are the performance targets based on?* A recorded benchmark environment and a fixed dataset, not a general guarantee.
54. *Did you copy an existing product?* Two existing tools were used as references. Reconciliation, lifecycle handling, stale protection, RBAC, multi-Agent coordination and the validation gate are not in either.

27. Open Decisions and Validation Gates

Item	Type	What must be decided	Why it matters	Who	When	Default
GST scope	Business decision	In or out for v1	Changes the voucher TDL and adds requirements	Business owner	Before voucher TDL is finalized	Out
Backup RPO/RTO	Business decision	Confirm ≤ 24 h / ≤ 4 h	Backup cost and design	Business owner	Before production	Proposed values
Journal entries in cash flow	Business decision	Include or exclude	Cash-flow completeness	Business owner	Before cash-flow module	Exclude
Fast-moving ranking basis	Business decision	Quantity or sales value	Quantity compares unlike units	Business owner	Before stock module	Quantity (value recommended)
Godown/batch stock, profitability	Business decision	Include in a later release?	Large additions	Business owner	After v1	Out
Validation gate G1-G29	Live Tally verification	Each item passed with evidence	Sync, classification, balances, aging, stock and attribution depend on them	Development team	During parser and sync design	Affected features disabled or flagged until passed
Agent packaging	Implementation test	Windows Service approach	Installation and updates	Development team	Agent build	NSSM/pywin32 service
Performance	Implementation	Confirm baseline	Valid performance	Development/QA	Before sign-off	Section 17.2

environment	test	is achievable	evidence			
Extraction batch defaults	Implementation test	Confirm 5,000 / 10,000	Avoid stalled exports	Development/QA	Before sign-off	5,000 / 10,000
Key-list check frequency	Implementation test	Every run or hourly	Cost on large companies	Development/QA	Sync build	Every run
Agent offline threshold, Tally uptime threshold	Implementation test	Confirm 5 minutes / 7 days	Useful alerts	Development team	Agent build	5 min / 7 days
Anomaly maximum-multiplier default	Implementation test	Choose a default	Rule currently inactive	Development team	Anomaly build	Not set
Separate Tally companies under one account	Future	Independent watermark streams	Only for consolidation	Architecture	Future release	Not supported
Automatic Agent failover	Future	Design if needed	Currently manual	Architecture	Future release	Manual

28. Implementation Readiness and Final Status

READY FOR DEVELOPMENT — Architecture and deployment; Agent registration, authentication, rotation, commands, routing, recovery, queue, targeting, batch limits and runtime handling; complete data model; synchronization (watermarks, lease, stale protection, batching, lifecycle, deletion detection, child replacement); accounting rules; reconciliation; aging and payment behaviour; stock classification; optional anomaly design; dashboard, drill-down and exports; RBAC and security; logging; failure modes; non-functional requirements; configuration; API; acceptance criteria AC-01 to AC-66.

VALIDATION REQUIRED BEFORE SYNC IS DECLARED COMPLETE — Gate items G1-G29 (Section 7).

IMPLEMENTATION TEST REQUIRED — Agent packaging, performance environment, extraction batch defaults, key-list frequency, Agent and Tally thresholds, anomaly multiplier default.

BUSINESS DECISION REQUIRED — GST scope, backup RPO/RTO, journal entries in cash flow, fast-moving ranking basis.

This document is **implementation-ready for development, subject to the validation gates above**. It is not claimed as production-ready.

Appendix A. Version History

Version	Summary
v1	Integration BRD/SRS: pushing and pulling data between an external app and Tally
v2-v3	Added analytics and AI features; natural-language query moved to an MCP-based design
v4	Review feedback applied: analytics made AI-free; TDL made the primary extraction mechanism; anomaly detection made optional via MCP + Claude
v5	Hard technical review: local Agent, GUID identity, voucher entries and bill allocations, reconciliation, RBAC, security, backup, acceptance criteria
v6	Accounting-direction rules, per-collection watermarks, stale protection, voucher replacement, command polling, onboarding, Not Yet Due aging, validation-gate approach
v7	Revenue attribution model, sign normalization, multiple Agents and routing, credential rotation without HMAC, allocation and return rules, master lifecycle, Never Sold, time zone and financial quarters
v7.1	Corrective addendum: reconciliation example corrected, customer and product attribution tightened, sync lease, command recovery
v7.2	Consolidated v7 and v7.1 into one document; schema placement and acceptance criteria completed
v7.3	Compared against tally-database-loader and Biz Analyst: group hierarchy, base voucher types, opening balances, stock snapshots, batch limits, company targeting, Tally runtime handling, faster deletion detection, custom fields, payment behaviour, responsive UI, scope decisions
v7.3 Complete Edition	This document: all versions merged into one self-contained specification, with no references to earlier versions for requirement text